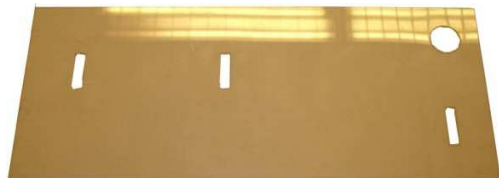
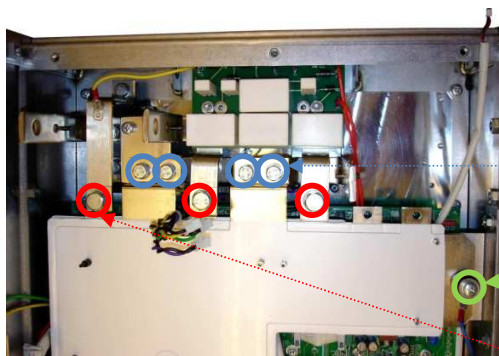


Plastic Parts Kit: VY1A1402



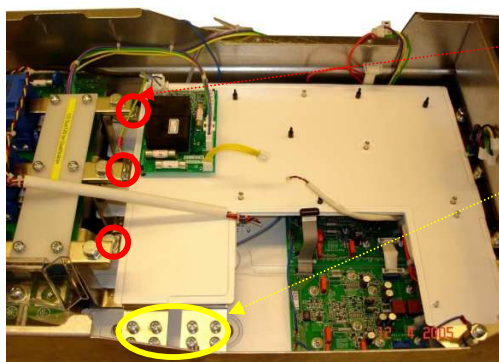
Don't forget to replace it during reassembling

DC Bus Bar KIT: VZ3N1322



Remove 4 screws (P3)

Remove 1 screw (P2)
&
Remove 1 wire:
Blue->X5 on braking unit board



Remove 6 screws (P1)

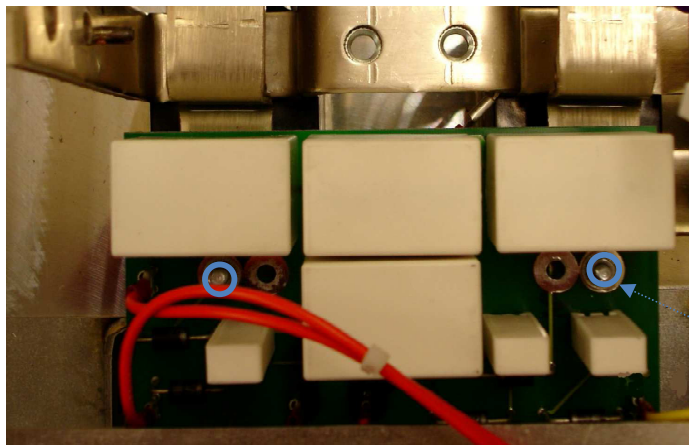
Remove 8 screws (P4)



Mark	Size	Torque
P1	M10x20	27Nm
P2	M6x12	5.5Nm
P3	M8x20	13.5Nm
P4	M8x20	13.5Nm

The DC bus bar kit can be removed.

Rectifier Snubber Circuit: VX4A1200



Disconnect 5 Wires right to left

Yellow X1 -> bus bar L1
 Red X4 -> bus bar PO
 Black X5 -> bus bar PC
 Red X10 -> RFS+X10 on Power board
 Red X11 -> RFS+X11 on Power board

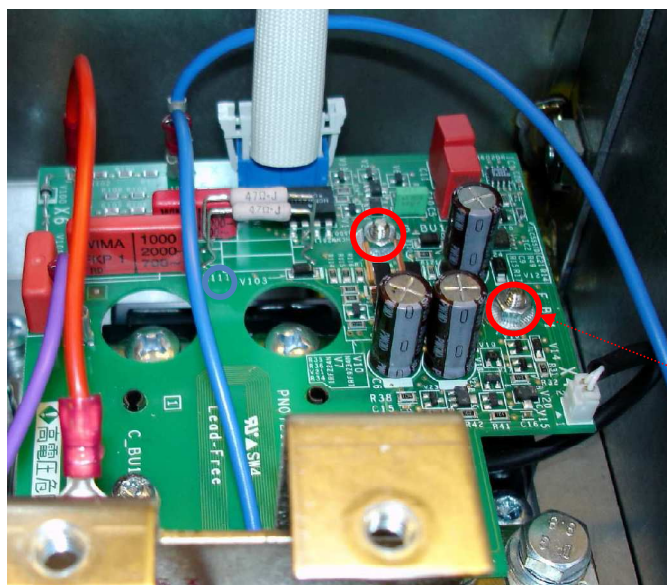
Remove 2 nuts.



Size	Torque
M4	1.2Nm

The Rectifier Snubber Circuit can be removed.

Braking unit Board: VZ3F1110



Disconnect 5 Wires right to left

Black X4 -> Thermal sensor B20
 Ribbon X1 -> X91 on Power board
 Blue X5 -> X52 on gate drive board IGBT W
 -> DC bus
 Purple X2 -> X93 on Power board
 X6 -> Bus bar PA

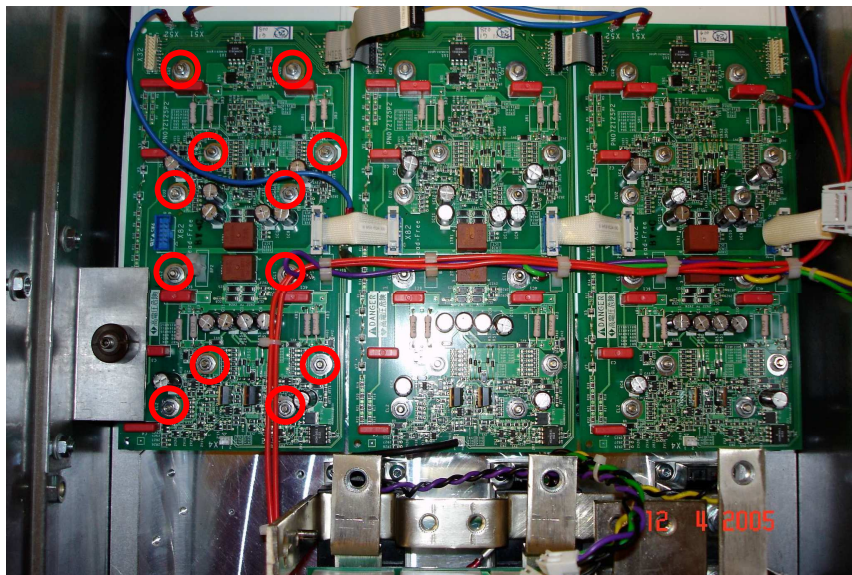
Remove 2 nuts.



Size	Torque
M4	1.2Nm

The Braking unit board can be removed.

Gate Drive Board IGBT: VX5A1201



For each branch U/V/W

Remove 12 nuts.

Disconnect the 14 wires.

Black X4 on gate drive Board IGBT V-> thermal sensor B1

Blue X51 on gate drive Board IGBT U->Power Interconnection bar W

Blue X52 on gate drive Board IGBT W->X5 on braking unit board

Blue X51<->X52 between GDB U/V/W

Yellow X2 on GDB U/Green X2 on GDB V/purple X2 on GDB W->X5 on power board

Ribbon X81 on GDB U ->X8 on Power board

Ribbon X81<-> X82 between GDB U/V/W

Ribbon X31 on GDB W ->X3 on Power board

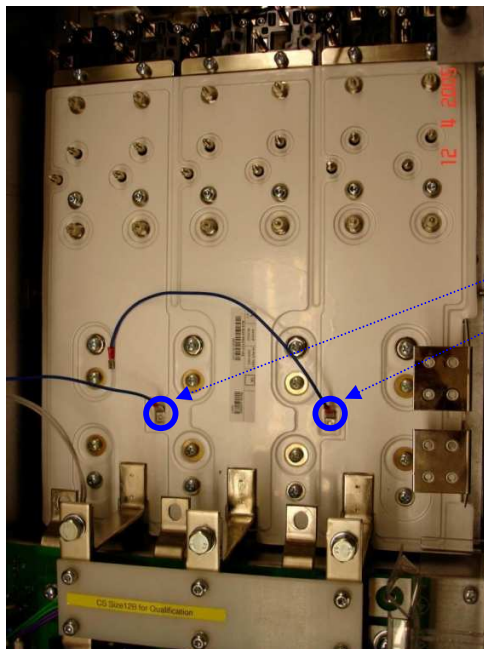
Ribbon X31<->X32 between GDB U/V/W

The Gate Drive Boards IGBT can be removed.



Size	Torque
M4	1.2Nm

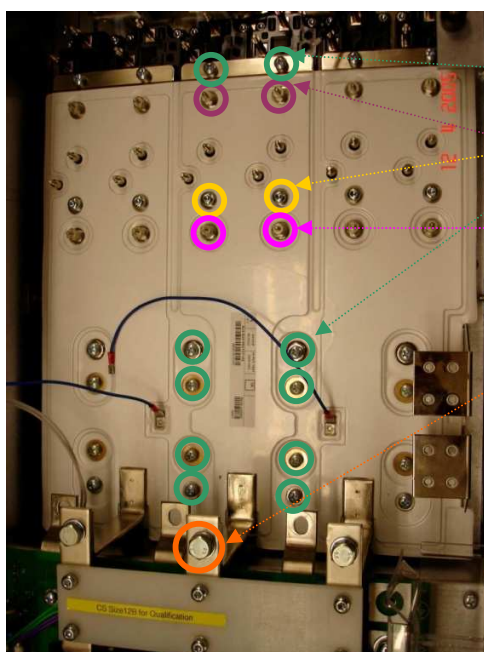
Bus Bar-Phase Size 13: VZ3N1322



Disconnect the 2 wires Left to right.

Blue wire on interconnection bar U -> UD0 on power board

Blue wire on interconnection bar W -> X51 on Gate Drive Board IGBT U



For each 3 AC bus bar, you should remove:

Remove 10 screws (P1-P2).

Remove 2 screws (P7)

Remove 2 screws (P6)

Remove 2 screws (P5)

Remove 1 screw (P8)

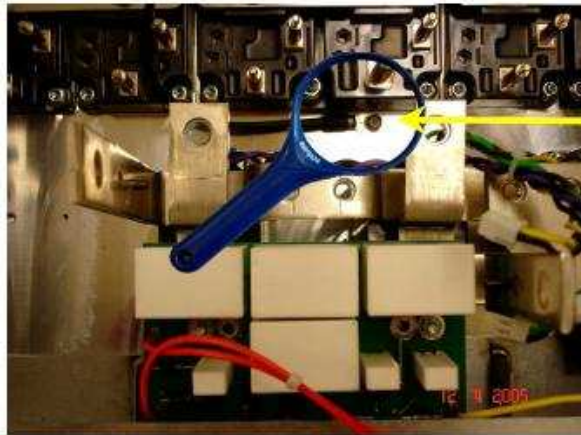
Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.



Mark	Size	Torque
P1-P2	M6x12	3.3Nm
P6	ST 10x6	3.3Nm
P7	M6x20	3.3Nm
P5	ST 10X11	3.3Nm
P8	M12x25	45Nm

The Bus Bar-Phase can be removed.

Thermal Sensor: VZ3G1104



Remove 1 screw.

Be careful, don't forget to connect the wire B1 on the connector "X4" of the Gate drive board IGBT V

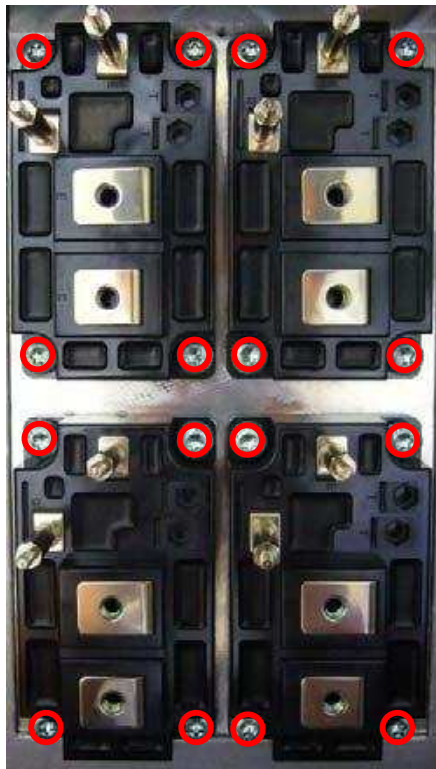
Be careful, don't forget to connect the second wire B20 on the connector "X4" of the braking unit board

The thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Module IGBT: VZ3IM1402M1271



For each 3 branch U/V/W, you should remove:

Remove 16 screws

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

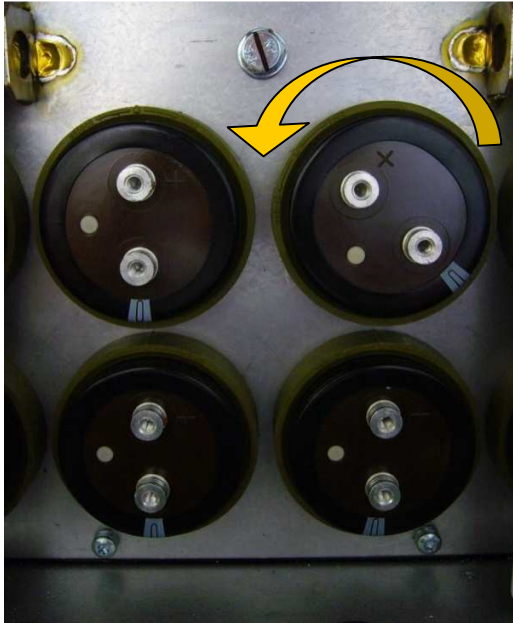
Don't forget applying the grease.

The Module IGBT can be removed.



Size	Torque
M6x20	3.0Nm

Lots of 8 capacitors: VY1ADC1113



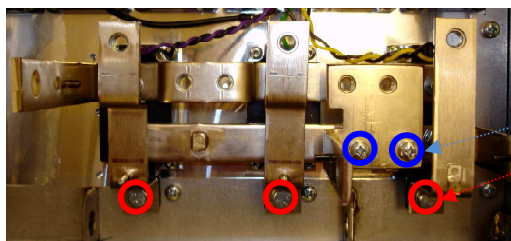
For all branch U/V/W, you should remove in total 8 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

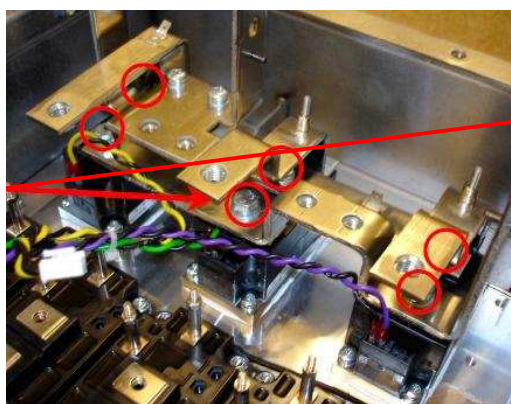
The Capacitors can be removed.

Rectifier module: VZ3TD1330M1601
AC Bus bar Kit: VZ3N1319



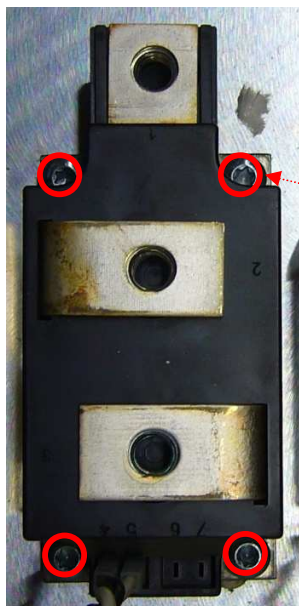
Step 1

Remove 2 screws (P1)
Remove 3 screws (P2)



Step 2

Remove 6 screws (P2)



Step 3

For each 3 branch L1/L2/L3

Remove the 4 screws (P3).

After changing, **be careful** to set up direction.

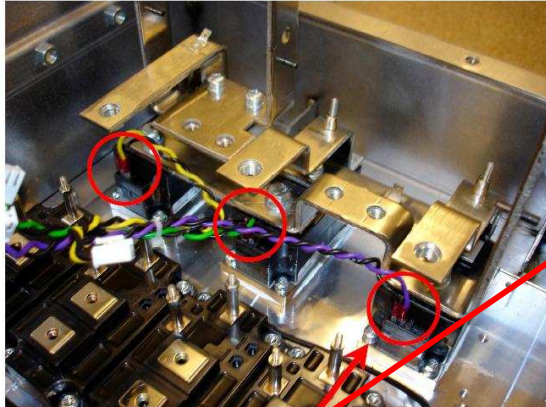
Don't forget applying the grease.

The rectifier can be removed.



Mark	Size	Torque
P2	M10x25	13.5Nm
P1	M6x12	5.5Nm
P3	M5x25	5.0Nm

Rectifier module thyristor: VZ3TD1330M1601

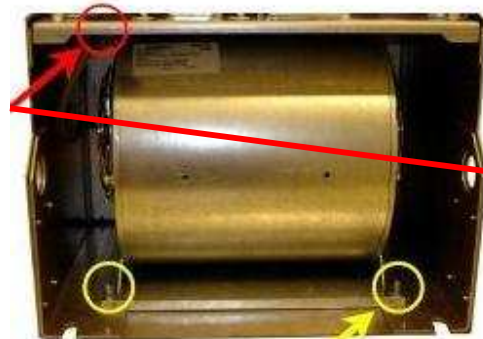


Be careful, after changing respect the colour line for the wires.

From left to right
[Black-Yellow]- [Black-Green] - [Black-Purple]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Fan Power Electronic: VZ3V1212



Be careful; don't forget to disconnect and reconnect the ground wire.

Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

Disconnect 1 connector

Fan turbine 1-> X3 on fan control board

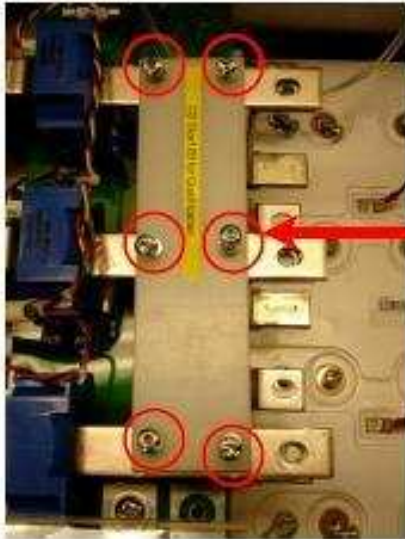
Remove 2 screws (J130)

The Turbine can be removed.



Mark	Size	Torque
J130	M6	5.5Nm

Plastic Parts Kit: VY1A1402



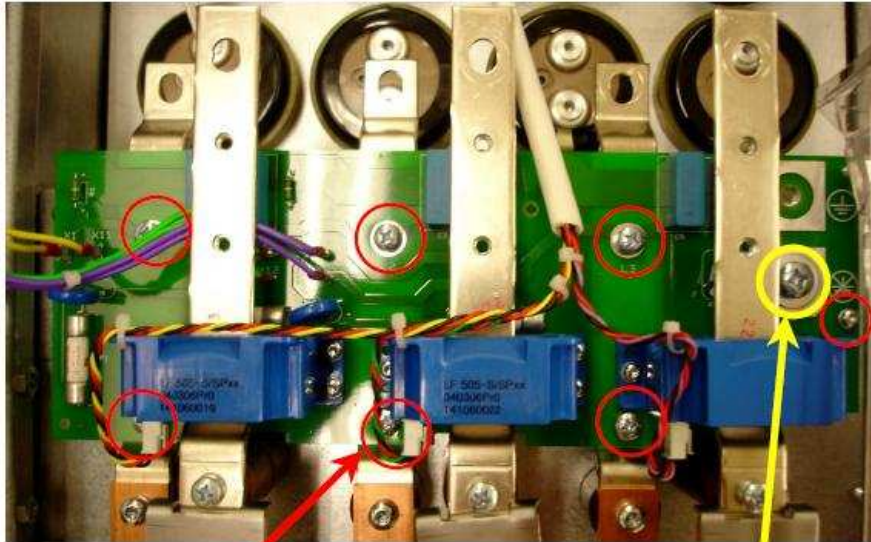
Remove 6 screws



Size	Torque
M6x20	5.5Nm

The Plastic Part can be removed.

RFI Filter Board: VX4A1114



Remove 1 screw (P2)

Remove 7 screws (P1)

The RFI filter board can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6x12	5.5Nm

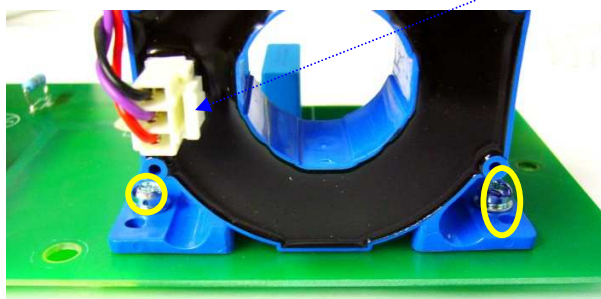
Motor Current Sensor: VY1A1108



For each 3 branch U/V/W, you should remove:

Remove 3 screws

Disconnect 1 wire
Current sensor->X11 on power board

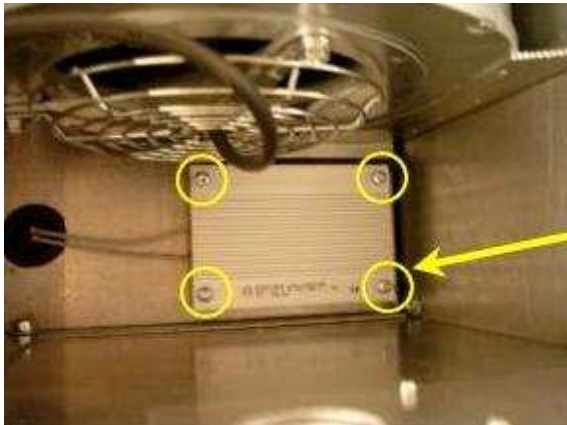


The Motor Current Sensor can be removed.



Size	Torque
M4x12	1.2Nm
M4	1.2Nm

Discharging resistor: VZ3R24KW125



Remove 4 screws



Size	Torque
M4x20	1.2Nm

The discharging resistor can be removed.



9.24.2 Product Assembling Drawing

No information

9.24.3 Product Cabling Drawing

Refer to following file: [Circuit Diagram Size 12.pdf](#)



9.25 ATV61/71 Size 13 (size, refer to 1.2)

9.25.1 Dismantling and reassembling

Size 13: ATV71HC20N4, ATV71HC25N4, ATV71HC28N4, ATV61HC25N4, ATV61HC31N4

ATV71HC20N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1425M1671	Rectifier Module Thyristor (425A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1321	AC Bus bar KIT
VZ3N1320	DC Bus Bar KIT
VZ3N1318	Wires KIT
VZ3IM1402M1271	Module IGBT pired (400A / 1200V)
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1109	Screws KIT
VY1ADC1114	Lots of 6 capacitors (3600µF / 400V)
VY1A1403	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1214	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC2025	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1201	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1115	RFI Filter Board
VW3A4522	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HC25N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1321	AC Bus bar KIT
VZ3N1320	DC Bus Bar KIT
VZ3N1318	Wires KIT
VZ3IM1602M1271	Lot of 2 Modules IGBT Paired (600A / 1200V)
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1109	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200µF / 400V)
VY1A1403	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1214	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC2531	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1202	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1115	RFI Filter Board
VW3A4520	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HC28N4

Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1321	AC Bus bar KIT
VZ3N1320	DC Bus Bar KIT
VZ3N1318	Wires KIT
VZ3IM1602M1271	Lot of 2 Modules IGBT Paired (600A / 1200V)
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1109	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200μF / 400V)
VY1A1403	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1214	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A71HC28N4	Power board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1202	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1115	RFI Filter Board
VW3A4520	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HC25N4

Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1425M1671	Rectifier Module Thyristor (425A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1321	AC Bus bar KIT
VZ3N1320	DC Bus Bar KIT
VZ3N1318	Wires KIT
VZ3IM1402M1271	Module IGBT pired (400A / 1200V)
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1109	Screws KIT
VY1ADC1114	Lots of 6 capacitors (3600μF / 400V)
VY1A1403	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1214	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC2025	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1201	Gate Drive Board
VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1115	RFI Filter Board
VW3A4522	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HC31N4

Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1321	AC Bus bar KIT
VZ3N1320	DC Bus Bar KIT
VZ3N1318	Wires KIT
VZ3IM1602M1271	Lot of 2 Modules IGBT Paired (600A / 1200V)
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1109	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200 μ F / 400V)
VY1A1403	Plastic Parts KIT
VY1A1302	Assembling KIT
VY1A1214	Front Cover with I/O Terminal Cover
VY1A1108	Motor Current Sensor
VX5A1HC2531	Power Board
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1202	Gate Drive Board
VX4A61101Y	Control bloc P $\geq$ 90kW
VX4A1200	Rectifier snubber circuit
VX4A1115	RFI Filter Board
VW3A4520	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O Terminal Cover:VY1A1214



Remove 9 screws.

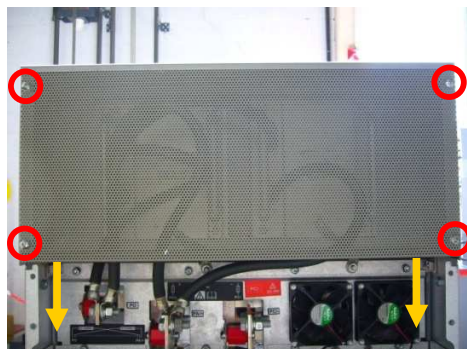


The front cover can be removed.



Size	Torque
M6x12	5.5Nm

Front Cover DC Choke: VW3A4522



Remove 4 screws.

The Self front cover can be removed.



Size	Torque
M6x12	5.5Nm

DC Choke connection: VW3A4520



Remove 4 nuts (P1).

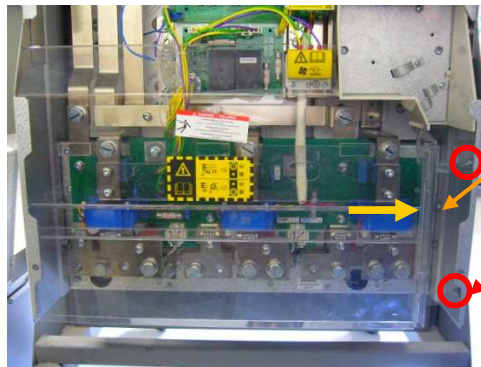
Remove 2 screws (P2)

The Self can be removed.



Mark	Size	Torque
P1	M8	13.5Nm
P2	M12x25	45Nm

Plastic Parts KIT: VY1A1403



Push here and disengage plastic part

Remove 2 nuts

The Plastic part KIT can be removed.



Size	Torque
M6	5.5Nm

Control bloc: VX4A61101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

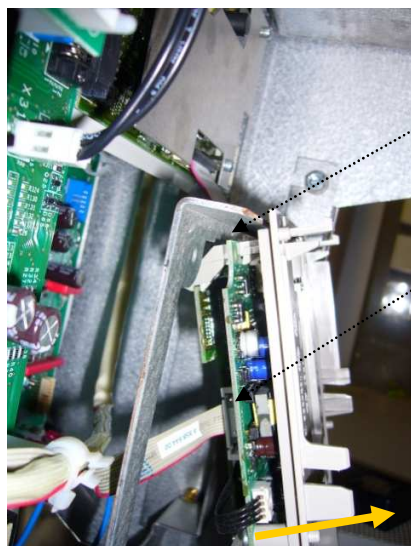


Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8).

Remove 1 screw (S38).



Disconnect the ribbon cable.
X3 from Control bloc "interface Board" -> X3 on Motor Control Board

Disconnect the ribbon cable.
X4 from Control bloc "interface Board" -> X4 on Power Board

The Control bloc can be removed.

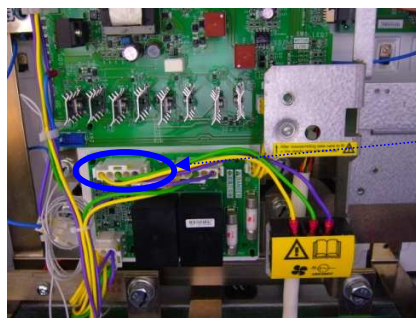


Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal Support for Control bloc



Remove 2 nuts.



Disconnect the wire:

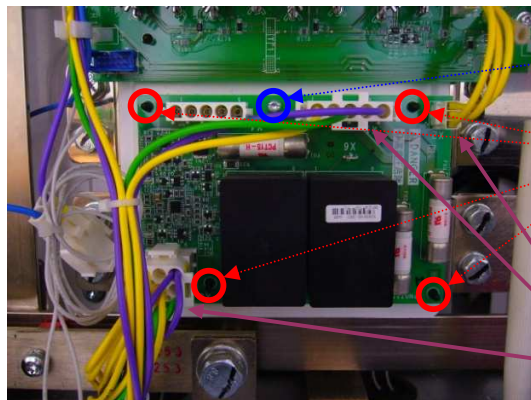
Yellow/Green/Purple -> X4 on fan control board

The metal support can be removed.



Size	Torque
M6	5.5Nm

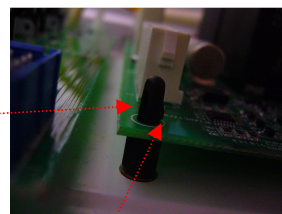
Fan Control Board: VX5A1400



Remove 1 screw.

Push on the 4 plastic supports to remove the board.

Disconnect the wires. From right to left:
Yellow X2->X14 on power board
Yellow/Green/Purple X1-> X1 Yellow X2 Green X3 Purple on filter card
Yellow/Green/Purple X3-> turbine alimentation



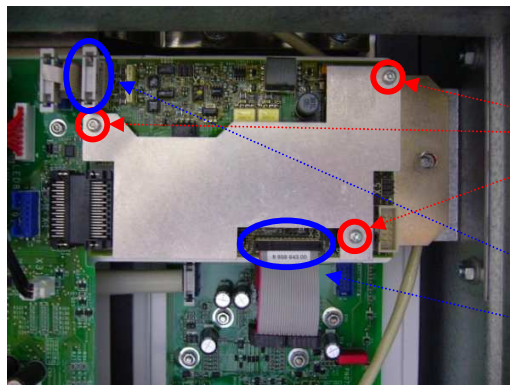
Push on the side.

The Fan control board can be removed.



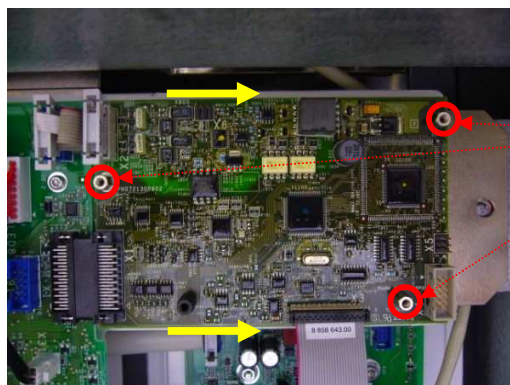
Size	Torque
M3x6	0.8Nm
M3x8	clips

Motor control Board: VX4A71101Y



Remove the 3 screws. (P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on Power Board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs. (P2)

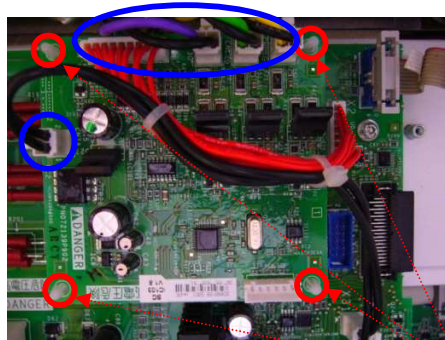
Be careful, at the time of reassembling;
don't forget to put the steel.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

The motor control board can be removed.

Soft Charge board: VX5A1300



Disconnect the 5 wires, *left to the right*:

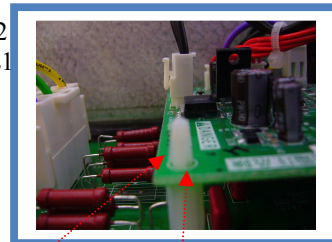
Black CN7A-> X31 on Power Board

Red CN2A-> X30 on Power Board

Purple/Black CNL3G ->Gate rectifier 3 on I3

Green/Black CNL2G ->Gate rectifier 2 on L2

Yellow/Black CNL1G -> Gate rectifier 1 on L1



Push on the 4 plastic supports
to remove the board.

Push on the side.

The Soft charge Board can be removed.



Size	Torque
M3x8	clips

Power board: VX5A1HC2531



Disconnect the 5 wires, *top to the down*:

Ribbon X2-> X2 on motor control board

Red X30-> CN2A on Soft charge board

Black X31-> CN7A on soft charge board

Yellow/Green/Purple X7-> X11 Yellow X12 Green X13 Purple on filter card

Yellow/Green/Purple X6->shunt on power board

Yellow/Green/Purple X5->alimentation Gate drive board IGBT Yellow/U Green/V Purple/W

Purple X93-> Purple X20->Blue X51 on Gate drive board IGBT U

Red RFS+X10 ->X10 on Rectifier Snubber Circuit.

White UD+R1 -> Discharging resistor

Ribbon X8->X8 on Command IGBT card

Red RFS+X11 ->X11 on Rectifier Snubber Circuit

Blue UD0 -> Power bus on U

White UD-R1 -> discharging resistor

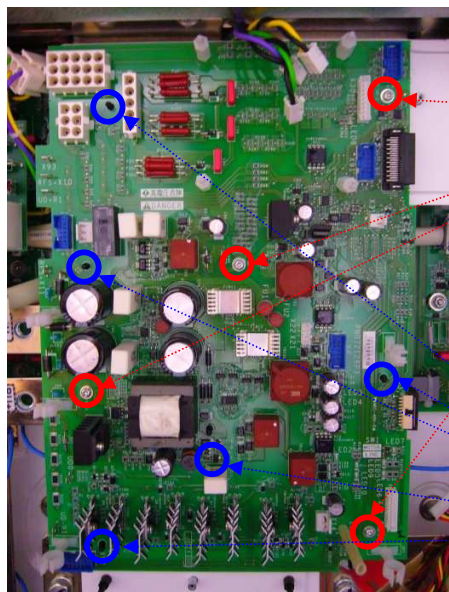
White X22 & X21-> two inside fan

Ribbon X4-> X4 on control block

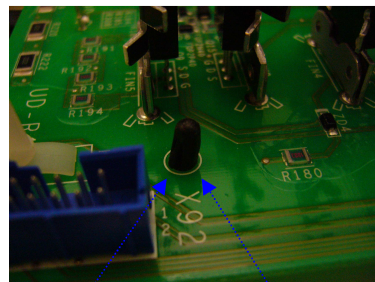
Ribbon X3 -> X31 on command IGBT board

Yellow X14 -> X2 on Fan control board

Multicolour X11-> currents sensors U V W



Remove the 4 screws. (S?)



Push on the 5 plastic supports to remove the board.

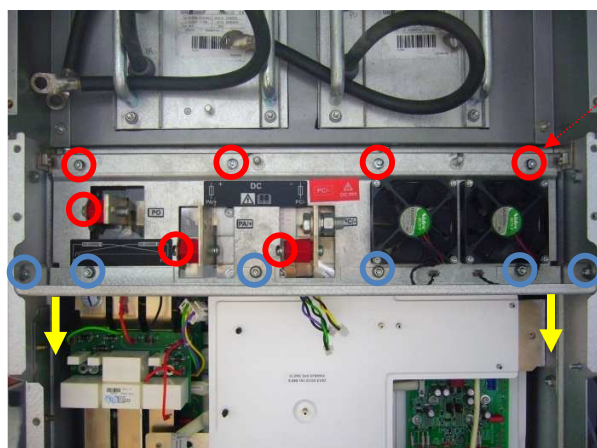
Push on the side.

The Power Board can be removed.



Size	Torque
M3x6	0.8Nm
M3x8	clips

Internal Fan 24 VDC: VZ3V1213



Remove 7 screws (P1).

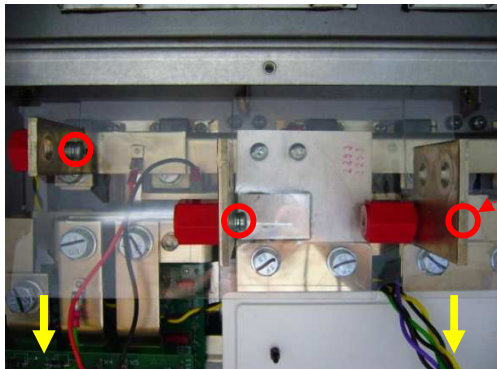
Remove 6 nuts (P2).



The Internal fan can be removed.

Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

Plastic Parts Kit: VY1A1403



Remove 3 screws.

Don't forget to replace it during reassembling

The plastic part can be removed.



Size	Torque
M6x12	5.5Nm

DC Bus Bar KIT: VZ3N1320



Remove 4 screws (P1)

Remove 1 screw (P2)

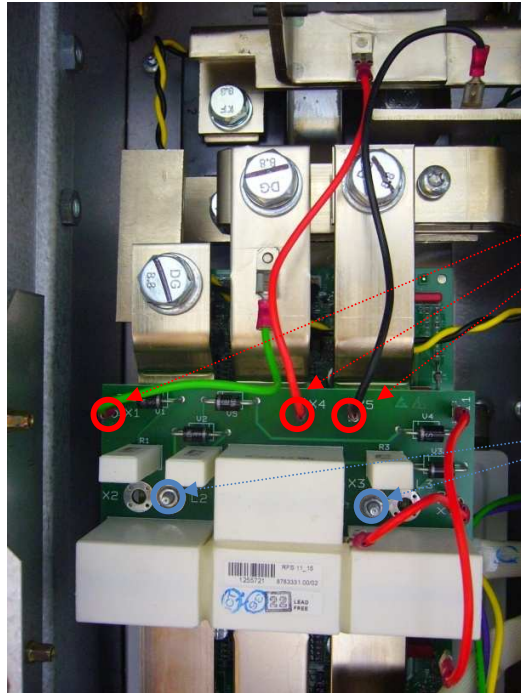
Remove 12 screws (P3)

The DC bus bar kit can be removed.



Mark	Size	Torque
P1	M10x20	27Nm
P2	M6x12	5.5Nm
P3	M8x20	13.5Nm

Rectifier Snubber Circuit: VX4A1200



Disconnect 3 Wires left to right

Green X1 -> bus bar L2
Red X4 -> bus bar PO
Black X5 -> bus bar PC

Remove 2 nuts.

The Rectifier Snubber Circuit can be removed.



Size	Torque
M4	1.2Nm

AC Bus bar KIT: VZ3N1321



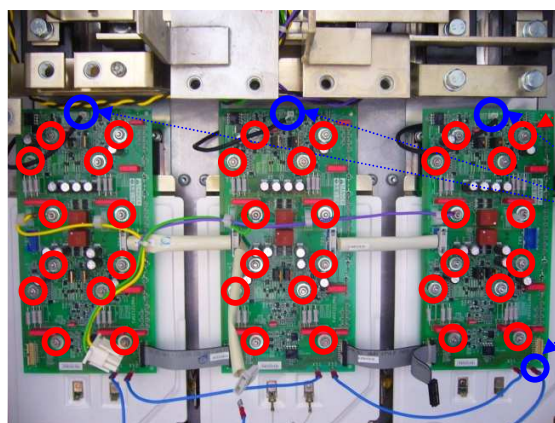
Remove 6 screws.

The AC Bus bar can be removed.



Size	Torque
M10x25	27Nm

Gate Drive Board IGBT: VX5A1202



Remove 36 nuts.

Disconnect the 4 wires top to down.

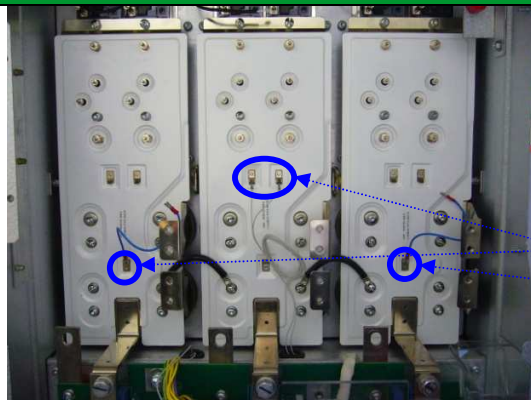
Black X4 on gate drive Board IGBT U/V/W-> thermal sensor
Blue X52 on gate drive Board IGBT W->Power Interconnection bar W

The Gate Drive Boards IGBT can be removed.



Size	Torque
M4	1.2Nm

Bus Bar-Phase Size 13: VZ3N1320

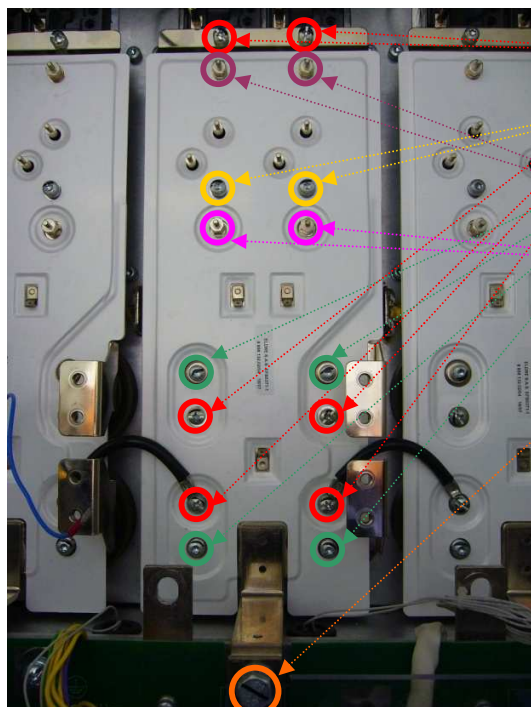


Disconnect the 4 wires Left to right.

Blue wire on interconnection bar U -> UD0 on power board

White wires on interconnection bar V -> discharging resistor

Blue wire on interconnection bar W -> X52 on Gate Drive Board IGBT W



For each 3 AC bus bar, you should remove:

Remove 6 screws (P1-P2).

Remove 2 screws (P7)

Remove 4 screws (P3-P4)

Remove 2 screws (P6)

Remove 2 screws (P5)

Remove 1 screw (P8)

Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.



Mark	Size	Torque
P1-P2	M6x12	3.3Nm
P6	ST 10x6	3.3Nm
P7	M6x20	3.3Nm
P5	ST 10X11	3.3Nm
P3-P4	M6x14	3.3Nm
P8	M12x25	45Nm

The Bus Bar-Phase can be removed.

Thermal Sensor: VZ3G1104



For each 3 branch U/V/W, you should remove:

Remove 1 screw.

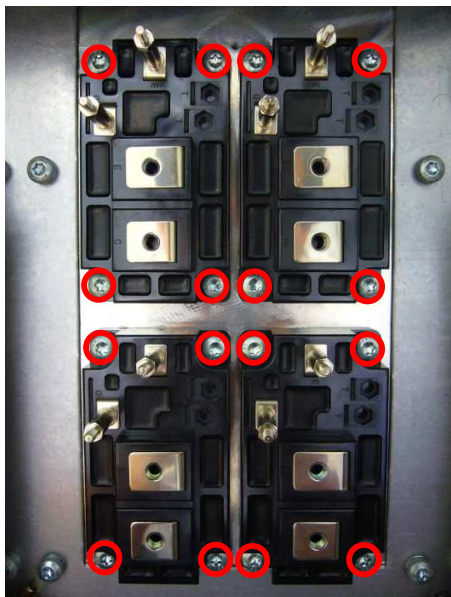
Be careful, don't forget to connect the wire on the connector "X4" of the Gate drive board IGBT

The thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Module IGBT. VZ3IM1602M1271



For each 3 branch U/V/W, you should remove:

Remove 16 screws

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

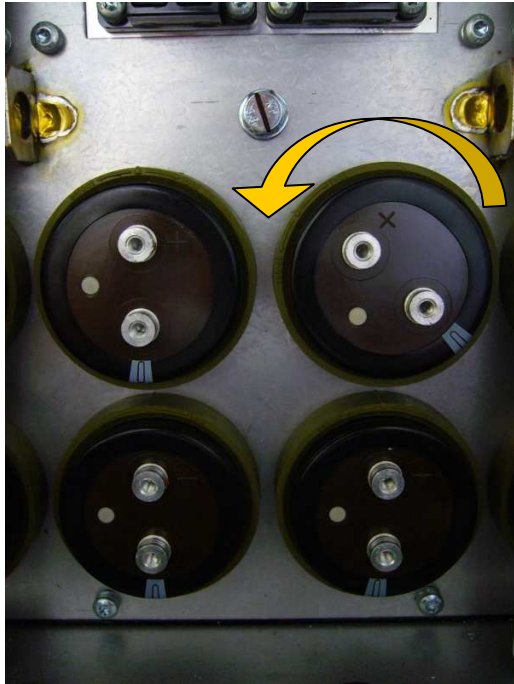
Don't forget applying the grease.

The Module IGBT can be removed.



Size	Torque
M6x20	3.0Nm

Lots of 6 capacitors: VY1ADC1112



For each 3 branch U/V/W, you should remove 4 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

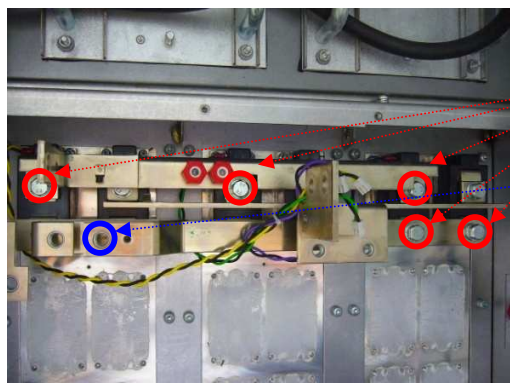
The Capacitors can be removed.

Rectifier module thyristor: VZ3TM1600M1671
Rectifier module diode: VZ3DM1600M1671
AC Bus bar Kit: VZ3N1321



Part 1

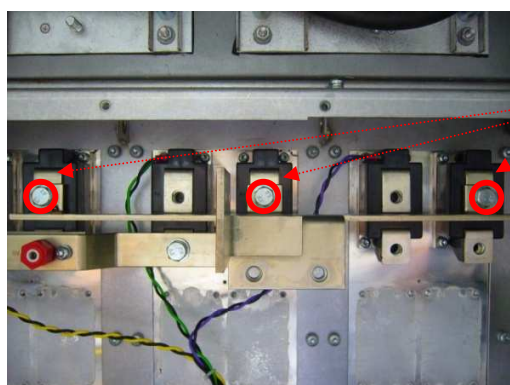
Remove 2 screws (P1)



Part 2

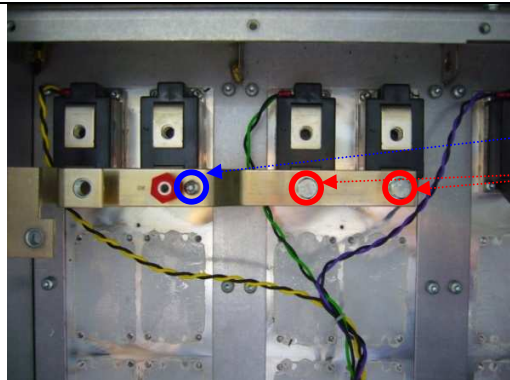
Remove 5 screws (P2)

Remove 1 screw (P1)



Part 3

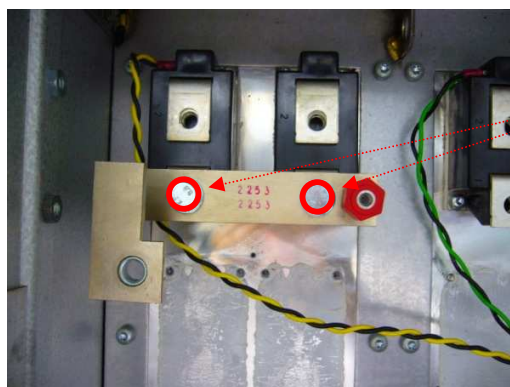
Remove 3 screws (P2)



Part 4

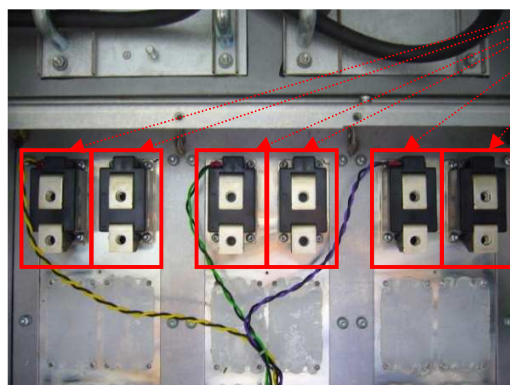
Remove 1 screw (P1)

Remove 2 screws (P2)



Part 5

Remove 2 screws (P2)



Remove the 24 screw(P3).

After changing, **be careful** to set up direction.

Don't forget applying the grease.



Mark	Size	Torque
P2	M10x25	13.5Nm
P1	M6x12	5.5Nm
P3	M5x25	5.0Nm

The rectifier can be removed.

Rectifier module thyristor: VZ3TM1600M1671 Rectifier module diode: VZ3DM1600M1671

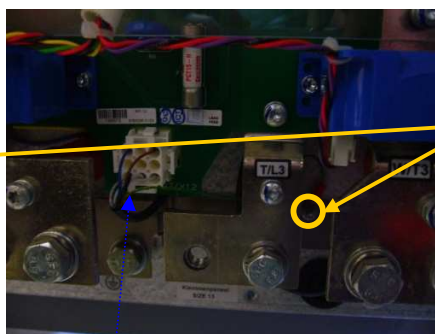
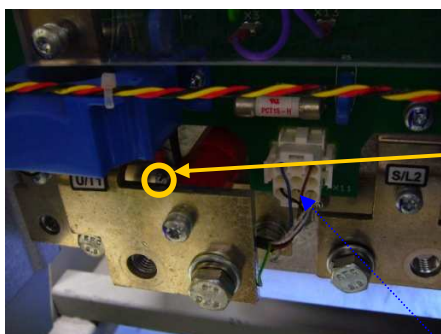


Be careful, after changing respect the colour line for the wires.

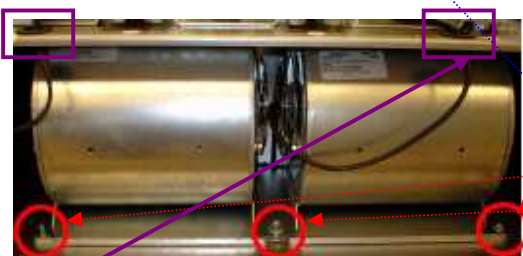
From left to right
[Yellow-Black]- [Green-Black] - [Purple-Black]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Fan Power Electronic. VZ3V1212



Be careful; don't forget to disconnect the ground wire.



Disconnect 2 connectors

Fan turbine 1->X11 on RFI filter board->X3 on fan control board
Fan turbine 2->X12 on RFI filter board->X3 on fan control board

Remove 2 screws (J130)
Remove 1 screw (J139)

Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

The Turbine can be removed.



Mark	Size	Torque
J130	M6	5.5Nm
J139	M8	13.5Nm

Plastic Parts Kit: VY1A1403



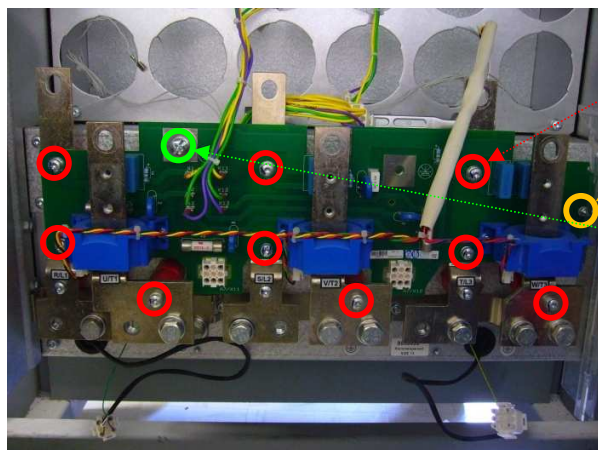
Remove 6 screws

The Plastic Part can be removed.



Size	Torque
M6x20	5.5Nm

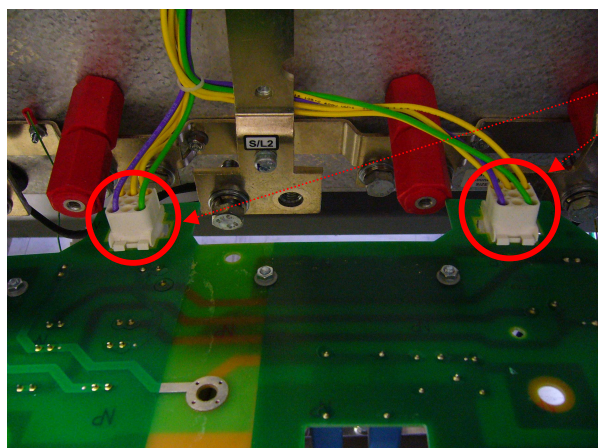
RFI Filter Board: VX4A1115



Remove 9 screws (P1)

Remove 1 nut (P2)

Remove 1 screw (P3)



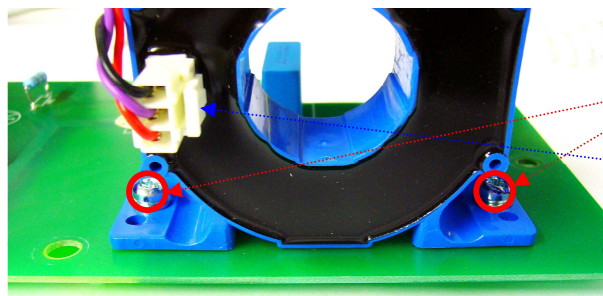
To change the board, don't forget to disconnect the 2 connectors X11 & X12

The RFI filter board can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M4	1.2Nm
P3	M6x12	5.5Nm

Motor Current Sensor: VY1A1108



For each 3 branch U/V/W, you should remove:

Remove 2 screws

Disconnect 1 wire

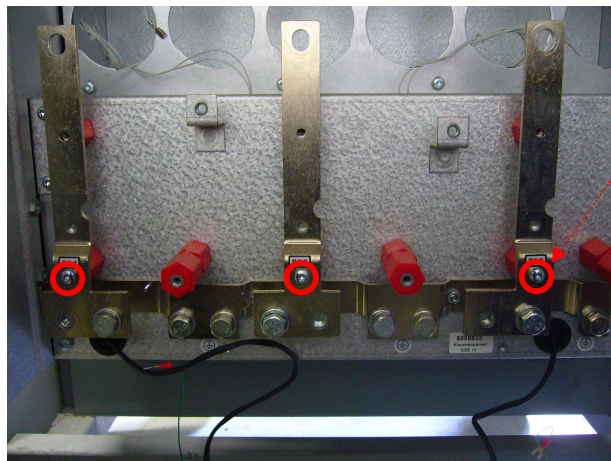
Current sensor->X11 on power board

The Motor Current Sensor can be removed.



Size	Torque
M4x12	1.2Nm
M4	1.2Nm

AC Bus bar Kit: VZ3N1321



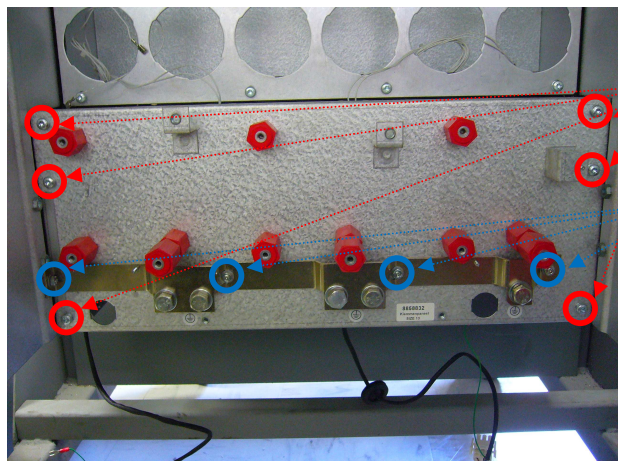
Remove 3 screws

The AC Bus bar can be removed.



Size	Torque
M6x14	5.5Nm

Earth Terminal Bar: VZ3N1321



Remove 6 screws

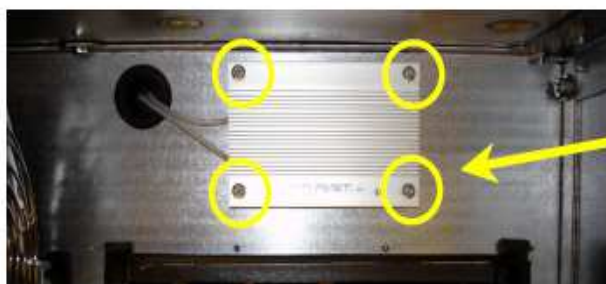
Remove 4 nuts



Size	Torque
M6x14	5.5Nm
M6	5.5Nm

The Earth Terminal Bar can be removed.

Discharging resistor: VZ3R24KW125



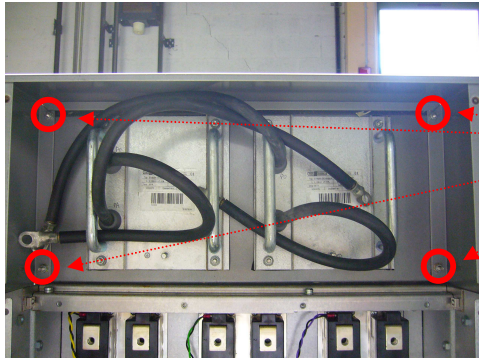
Remove 4 screws



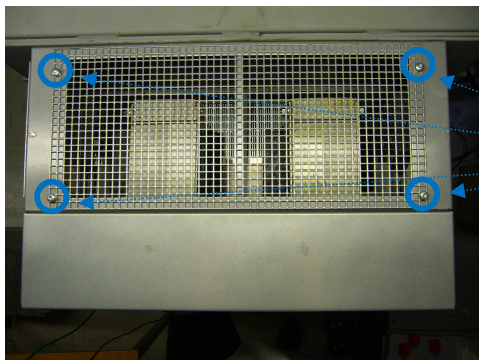
Size	Torque
M4x20	1.2Nm

The discharging resistor can be removed.

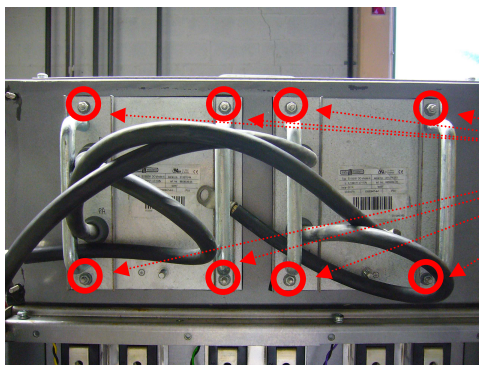
DC Choke: VW3A4520



Remove the 4 nuts



Remove the 4 screws



Remove the 8 nuts

The DC Choke can be removed.



Size	Torque
M6	5.5Nm
M6x12	5.5Nm



9.25.2 Product Assembling Drawing

No information

9.25.3 Product Cabling Drawing

Refer to following file: [Cabling diagram size 13.pdf](#)



9.26 ATV71-61 Size F14 (size, refer to 1.2)

9.26.1 Dismantling and reassembling

Frame 14: ATV71HC31N4, ATV71HC40N4, ATV61HC40N4, ATV61HC50N4

ATV71HC31N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1333	Wires KIT
VZ3N1331	AC Bus Bar KIT
VZ3N1329	DC Bus Bar KIT
VZ3IM1603M1271	3 Modules pired IGBT 600A/1200V
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1110	Screws KIT
VY1ADC1114	Lots of 6 capacitors (3600μF / 400V)
VY1A1406	Plastic Parts KIT
VY1A1305	Assembling KIT
VY1A1216	Front Cover with I/O Terminal Cover
VY1A1109	Motor Current Sensor
VX5A1HC3140	Power Board
VZ3F1113	Braking Unit KIT
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1203	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1118	RFI Filter Board
VW3A4522	DC Choke
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV71HC40N4	
Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1425M1671	Rectifier Module Thyristor (425A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1340	AC Bus Bar KIT
VZ3N1339	DC Bus Bar KIT
VZ3N1333	Wires KIT
VZ3IM1603M1271	3 Modules pired IGBT 600A/1200V
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1110	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200μF / 400V)
VY1A1406	Plastic Parts KIT
VY1A1305	Assembling KIT
VY1A1216	Front Cover with I/O Terminal Cover
VY1A1109	Motor Current Sensor
VX5A1HC4050	Power Board
VZ3F1113	Braking Unit KIT
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1203	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1117	RFI Filter Board
VW3A4523	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

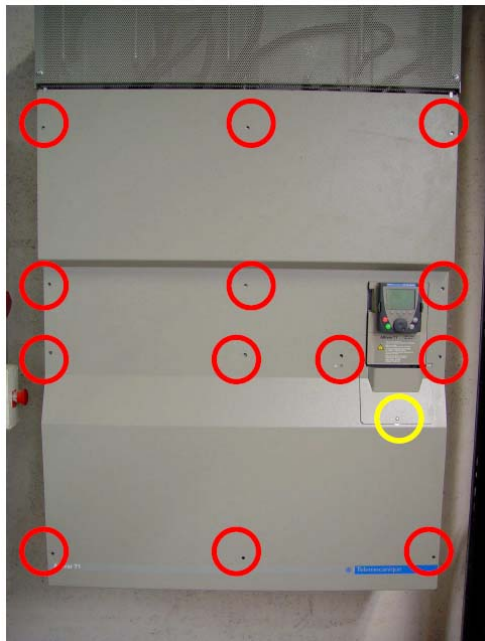
ATV61HC40N4

Reference	Designation
VZ3V1213	Internal Fan
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1333	Wires KIT
VZ3N1331	AC Bus Bar KIT
VZ3N1329	DC Bus Bar KIT
VZ3IM1603M1271	3 Modules IGBT apairés 600A/1200V
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1110	Screws KIT
VY1ADC1114	Lots of 6 capacitors (3600µF / 400V)
VY1A1406	Plastic Parts KIT
VY1A1305	Assembling KIT
VY1A1216	Front Cover with I/O Terminal Cover
VY1A1109	Motor Current Sensor
VX5A1HC3140	Power Board
VZ3F1113	Braking Unit KIT
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1203	Gate Drive Board
VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1118	RFI Filter Board
VW3A4522	DC Choke
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

ATV61HC50N4

Reference	Designation
VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic
VZ3TM1425M1671	Rectifier Module Thyristor (425A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1340	AC Bus Bar KIT
VZ3N1339	DC Bus Bar KIT
VZ3N1333	Wires KIT
VZ3IM1603M1271	3 Modules IGBT apairés 600A/1200V
VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1110	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200µF / 400V)
VY1A1406	Plastic Parts KIT
VY1A1305	Assembling KIT
VY1A1216	Front Cover with I/O Terminal Cover
VY1A1109	Motor Current Sensor
VX5A1HC4050	Power Board
VZ3F1113	Braking Unit KIT
VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board
VX5A1203	Gate Drive Board
VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit
VX4A1117	RFI Filter Board
VW3A4523	DC Choke
VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O terminal cover: VY1A1216



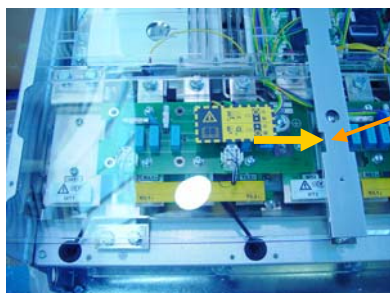
Remove 17 screws

The Front Cover can be removed



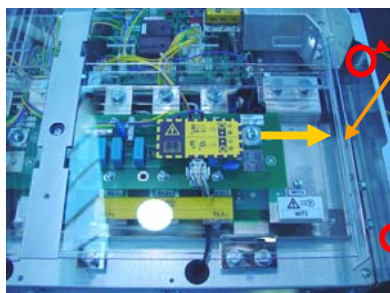
Size	Torque
M6x12	5.5Nm

Plastic Part Kit: VY1A1406



Push here and disengage plastic part

Remove 2 screws

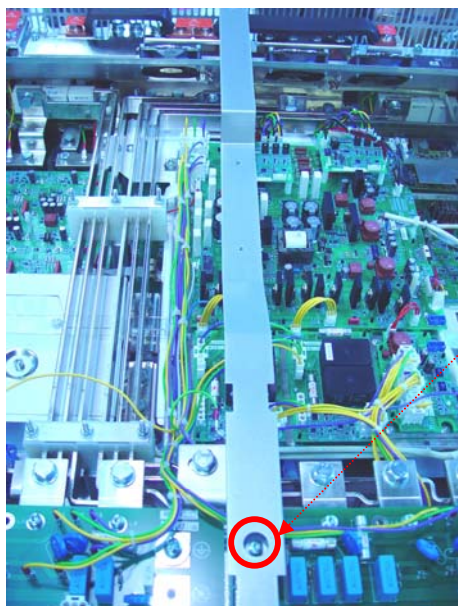


The Plastic part kit can be removed.

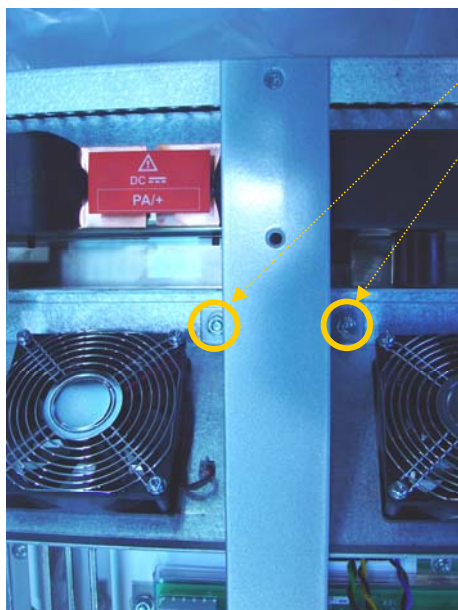


Size	Torque
M6	5.5Nm

Metal kit.



Remove 1 screw (P1)



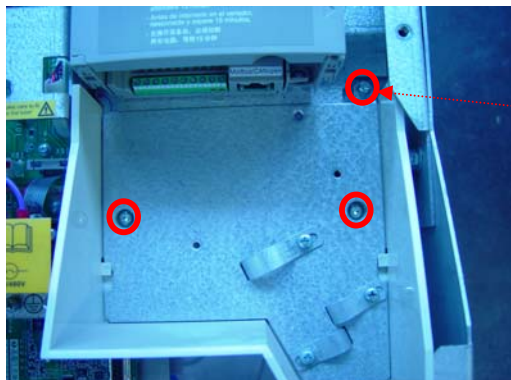
Remove 2 nuts (P2)

The metal part can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

Plastic Part Kit: VY1A1406



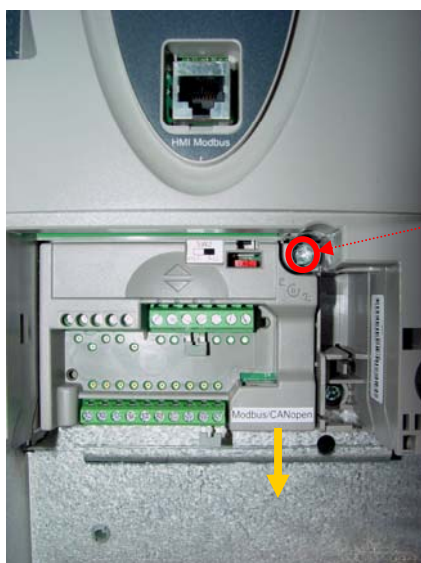
Remove 3 screws

The Plastic Part kit can be removed.



Size	Torque
M4x12	1.2Nm

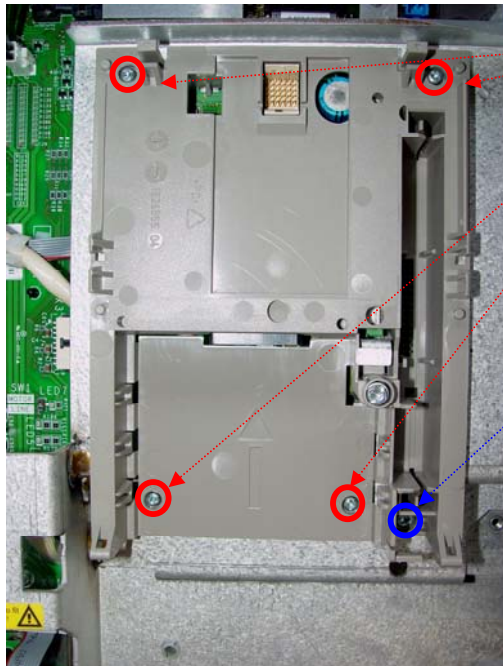
Control bloc: VX4A71101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.

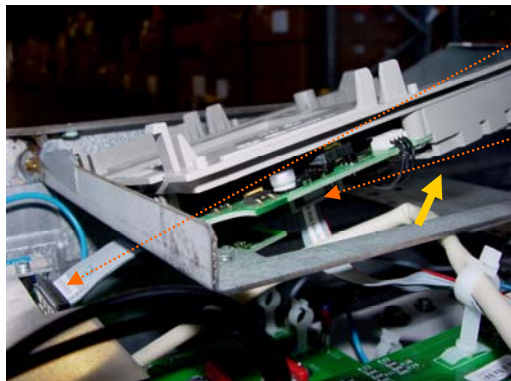


Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8).

Remove 1 screw (S38).



Disconnect the ribbon cable.

X3 from Control bloc "interface Board"->X3 on Motor Control Board

Disconnect the ribbon cable.

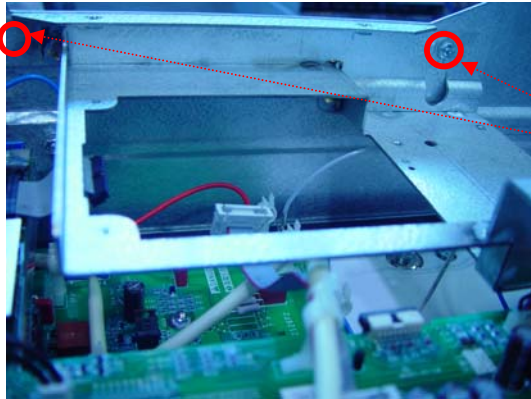
X4 from Control bloc" interface Board"->X4 on Power Board

The Control bloc can be removed.

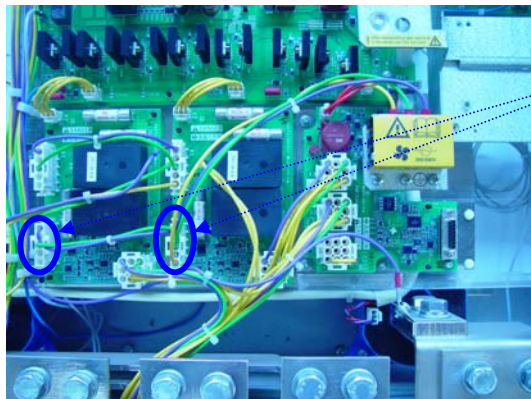


Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal support for control bloc



Remove 2 nuts



Disconnect wire:

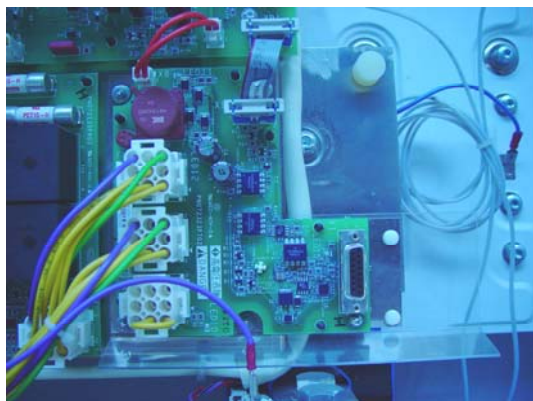
Yellow/Green/Purple TB1 -> X4 on both fan control board

The metal kit can be removed.



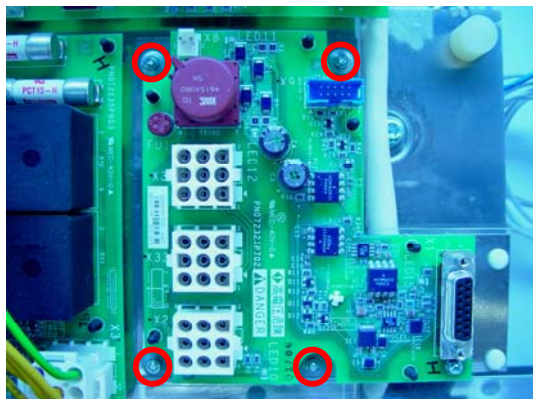
Size	Torque
M6	5.5Nm

Braking Unit KIT: VZ3F1113

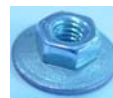


Disconnect 5 wires top to down:

- Red X8 -> X81 on Power board
- Ribbon X91 -> X91 on Power board
- Yellow/Green/Purple X3 -> X3 on fan control board 2
- Yellow/Green/Purple X33 -> X13 on RFI filter board 2
- Yellow X2 -> Shunt X2 on braking unit device



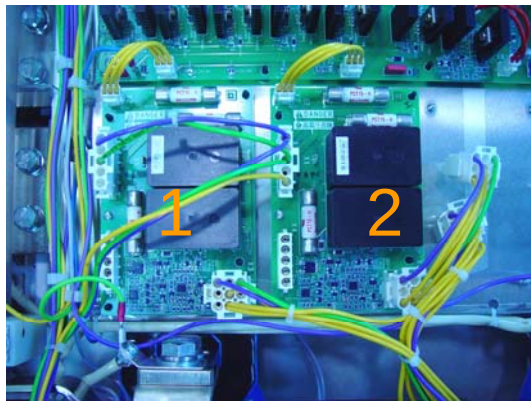
Remove 4 nuts



Size	Torque
M3	0.8Nm

The Braking Unit KIT can be removed.

Fan Control Board. VX5A1400



Disconnect wires. From top to down:

Fan control Board 1

Yellow X2->X14 on Power board

Yellow/Green/Purple X1->X1 on fan control Board 2

Yellow/Green/Purple X3-> X11 & X12 on RFI filter board

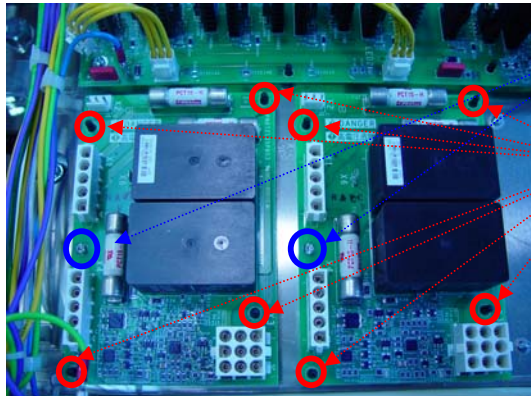
Fan control Board 2

Yellow X2->X15 on Power board

Yellow/Green/Purple X1->X1 on Fan control board 1

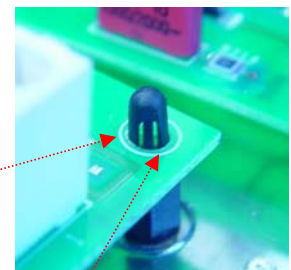
Yellow X1/Green X2/ Purple X3 on RFI filter board

Yellow/Green/Purple X3-> X3 on braking unit device



Remove 2 screws

Push on the 8 plastic supports to remove the board.



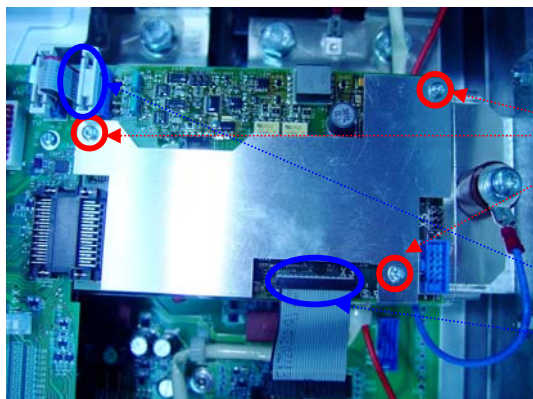
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

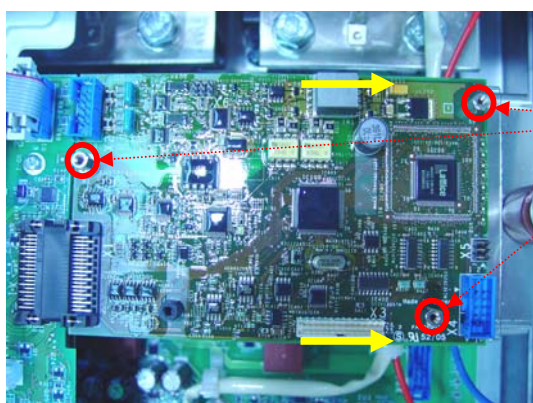
The Fan control board can be removed.

Motor control Board: VX4A71101Y



Remove the 3 screws.(P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on Power board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs.(P2)

Be careful, at the time of reassembling;
don't forget to put the steel.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

The motor control board can be removed.

Soft Charge board: VX5A1300



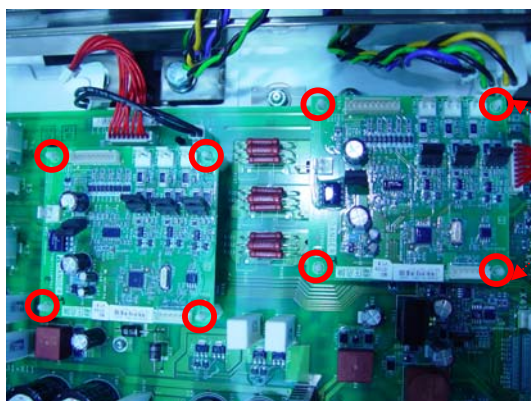
Disconnect wires, *left to the right*:

Soft charge Board 1

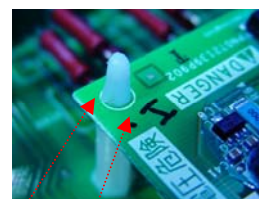
Blue CN7A-> X31 on Power board
 Red CN2A-> X30 on power board
 Purple/Black CNL3G ->Gate rectifier 3.1 on L3.1
 Green/Black CNL2G ->Gate rectifier 2.1 on L2.1
 Yellow/Black CNL1G -> Gate rectifier 1.1 on L1.1

Soft charge Board 2

Blue CN7A-> X41 on Power board
 Red CN2A-> X40 on power board
 Purple/Black CNL3G ->Gate rectifier 3.2 on L3.2
 Green/ Black CNL2G ->Gate rectifier 2.2 on L2.2
 Yellow/ Black CNL1G -> Gate rectifier 1.2 on L1.2



Push on the 8 plastic supports
to remove the board.



Push on the side.



Size	Torque
M3x8	clips

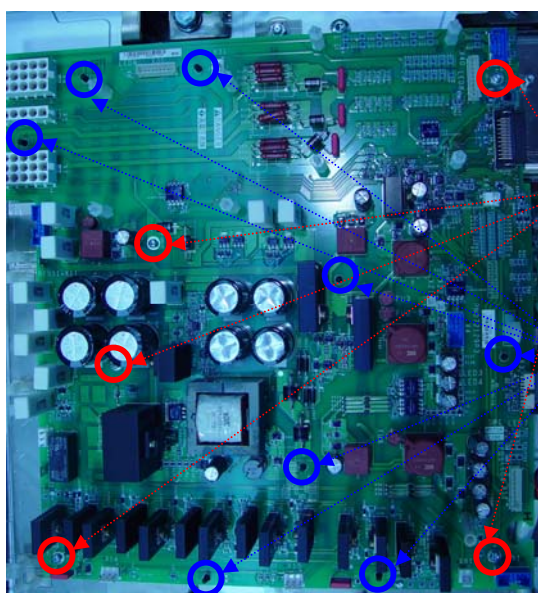
The Soft charge Board can be removed.

Power Board: VX5A1HC4050



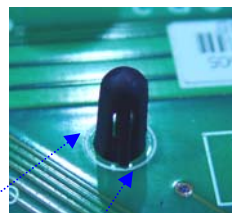
Disconnect the 21 wires, *unclockwise*:

Black X41-> CN7A on soft charge board 2
 Red X40-> CN2A on soft charge board 2
 Ribbon X2-> X2 on motor control board
 Black X31-> CN7A on soft charge board 1
 Red X30-> CN2A on Soft charge board 1
 Yellow/Green/Purple X6-> X6 shunt on Power Board
 Yellow/Green/Purple X5-> bus bar U Yellow, bus bar V Green, bus bar W Purple
 Yellow/Green/Purple X7->Yellow X11/Green X12/Purple X13 on both RFI filter Board
 Ribbon X8 -> X82 on Gate drive Board IGBT U
 Red RFS31+X11 -> X11 on Snubber circuit 1
 White UD-R1 -> discharging resistor
 Blue UD0 -> interconnection bus bar W
 Yellow X14-> X2 on Fan control Board 1
 Yellow X15 -> X2 on Fan control board 2
 Red X81-> X8 on Braking unit device
 Ribbon X91-> X91 on Braking unit device
 Multicolour X11-> currents sensors U V W
 Ribbon X3-> X32 on Gate drive board W
 Ribbon X4-> X4 on control block
 White X22-> inside Fan alimentation
 White X21-> inside Fan alimentation



Remove the 5 screws.

Push on the 8 plastic supports to remove the board



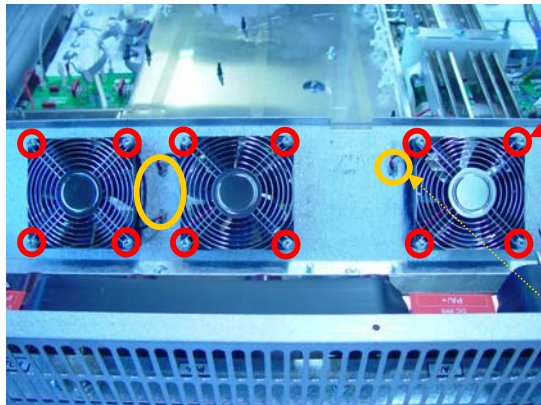
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

The Power board can be removed.

Internal fan kit: VZ3V1213



Remove 16 screws.

Disconnect 4 Wires

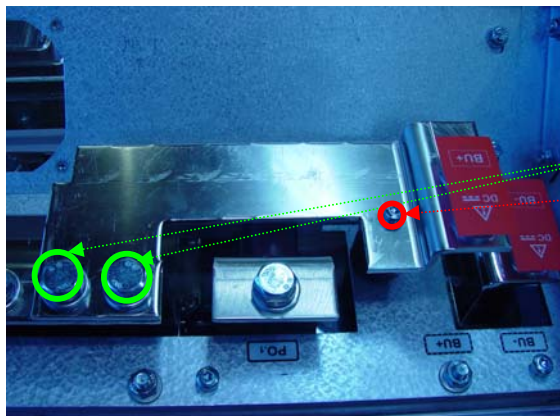
Black Fan-> X21 X22 on Power board

The Internal Fan can be removed.



Size	Torque
M4x35	1.2Nm

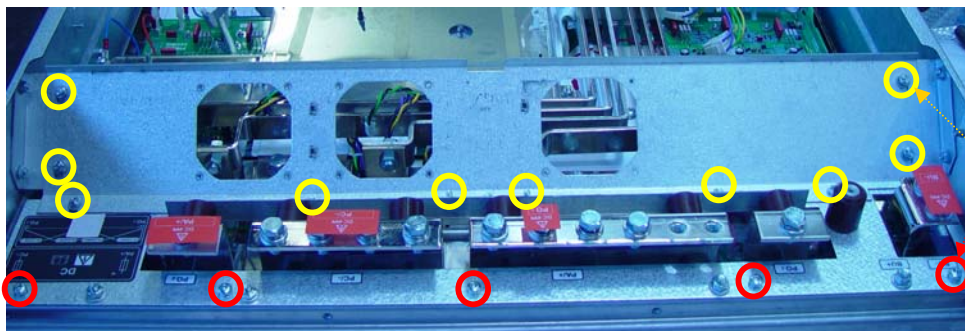
AC Bus Bar KIT: VZ3N1340



Step 1

Remove 2 screws (P3).

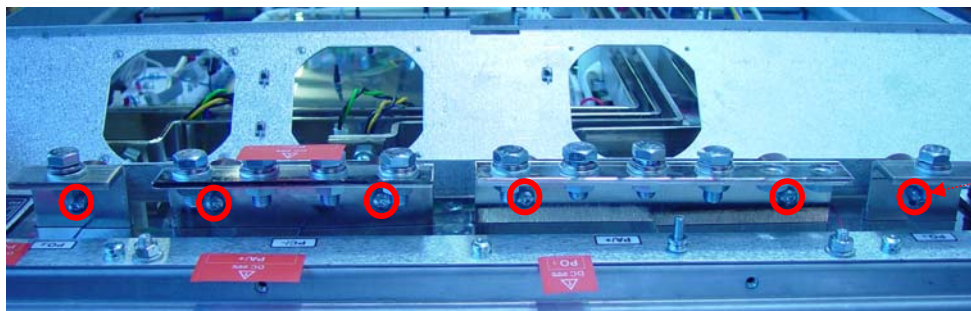
Remove 1 screw (P1)



Step 2

Remove 10 nuts (P2).

Remove 5 screws (P1)



Step 3

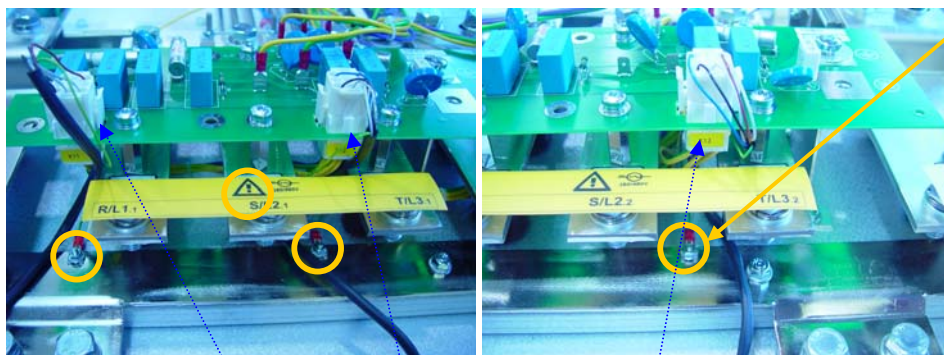
Remove 6 screws (P1).



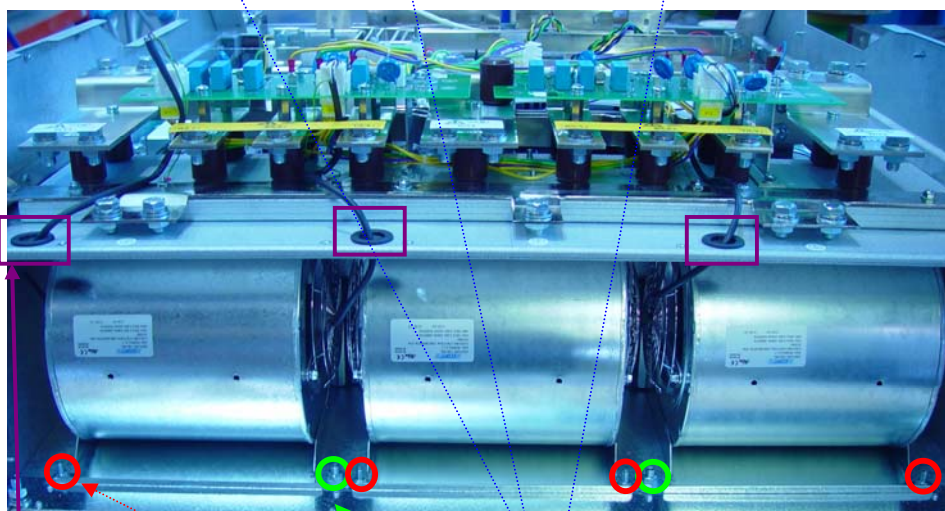
Mark	Size	Torque
P3	M12x25	45Nm
P1	M6x14	5.5Nm
P2	M6	5.5Nm

The AC Bus bar kit can be removed.

Fan Power Electronic. VZ3V1212



Be careful; don't forget to disconnect the ground wire.



Disconnect 3 connectors
Fan turbine 1->X11 on RFI filter board 1->X3 on fan control board 1
Fan turbine 2->X12 on RFI filter board 1->X3 on fan control board 1
Fan turbine 3->X13 on RFI filter board 2->X33 on Braking unit device

Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

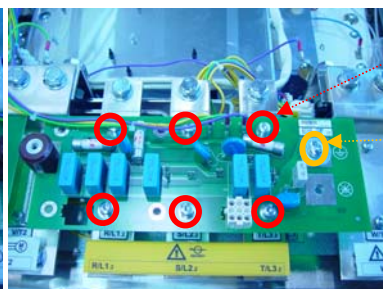
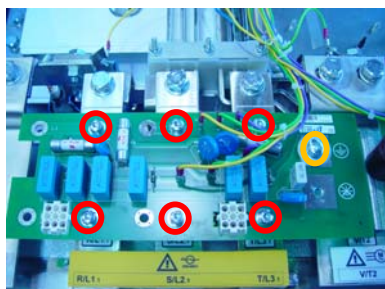
Remove 4 screws (J130)
Remove 2 screws (J139)

The Turbine can be removed.



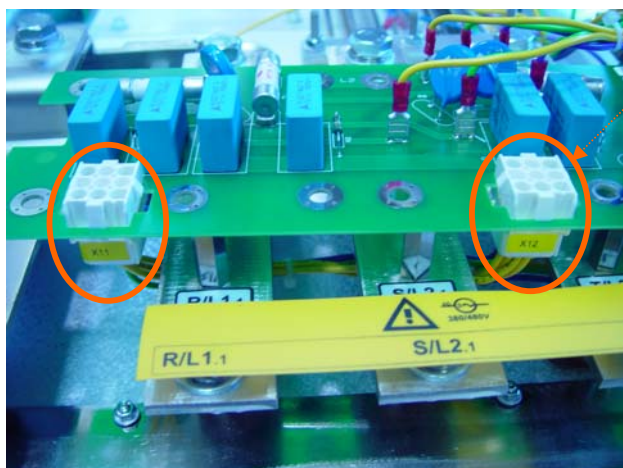
Mark	Size	Torque
J130	M6	5.5Nm
J139	M8	13.5Nm

RFI Filter Board. VX4A1117



Remove 12 screws (P1)

Remove 2 screws (P2)



To change the board, don't forget to disconnect the 3 connectors

X11 X12 on RFI Filter Board 1

And

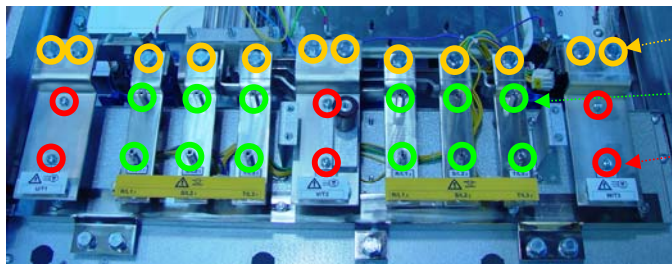
X13 on RFI Filter Board 2

The Both RFI filter board can be removed.



Mark	Size	Torque
P2	M6x12	5.5Nm
P1	M6x14	5.5Nm

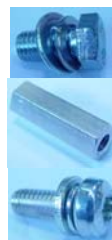
AC Bus Bar KIT: VZ3N1340



Remove 12 screws (P1)

Remove 12 screws (P2)

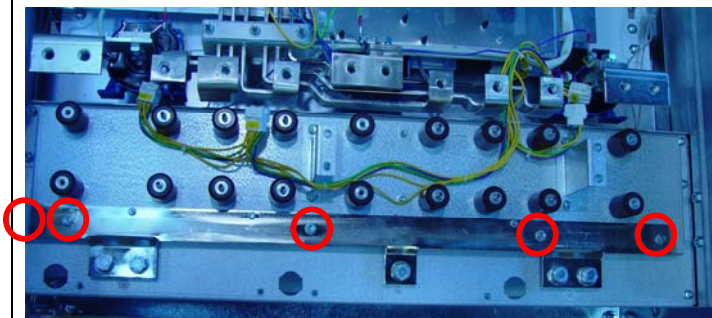
Remove 6 screws (P3)



Mark	Size	Torque
P1	M12x25	45Nm
P2	M6x22	5.5Nm
P3	M6x14	5.5Nm

The AC Bus Bar kit can be removed.

Earth terminal Bar: VZ3N1332



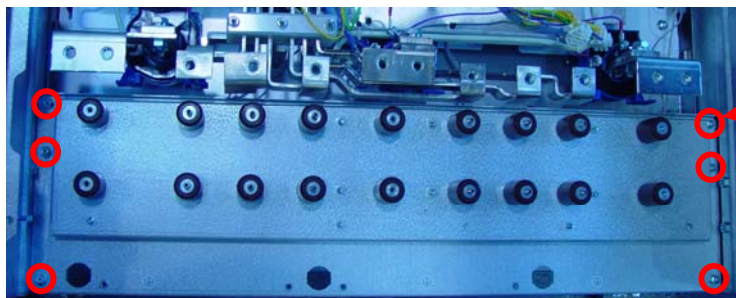
Remove 5 nuts



Size	Torque
M6	5.5Nm

The Earth terminal bar can be removed.

Metal Body Part KIT



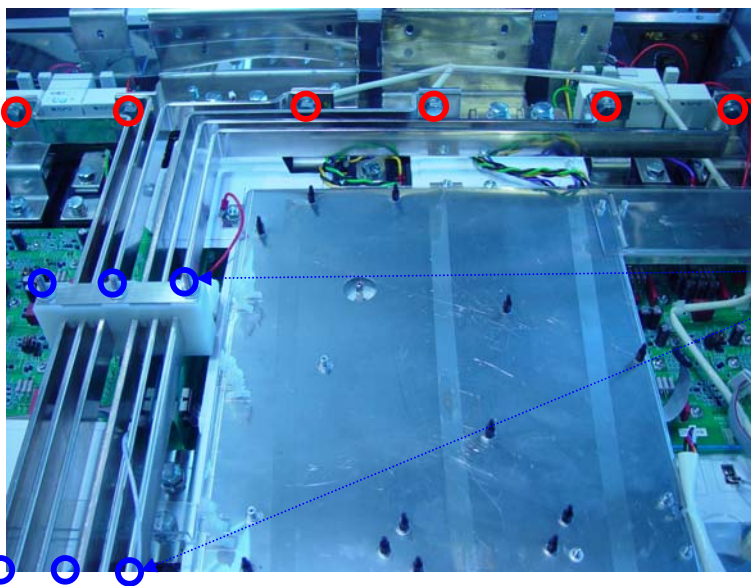
Remove 6 screws

The Metal Body Part kit can be removed.



Size	Torque
M6x14	5.5Nm

AC Bus Bar KIT: VZ3N1340



Remove 6 screws

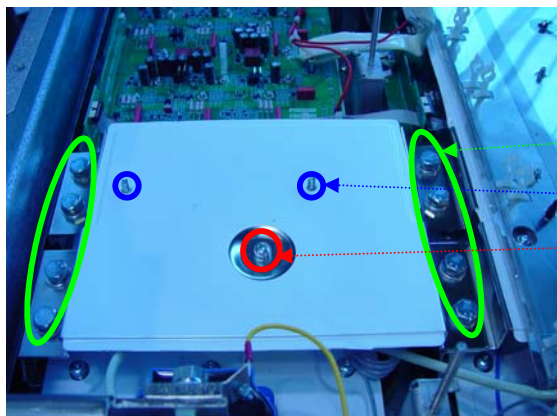
Remove 6 nuts

The AC Bus bar kit can be removed.



Size	Torque
M12x25	45Nm
M6	5.5Nm

DC bus bar kit: VZ3N1339

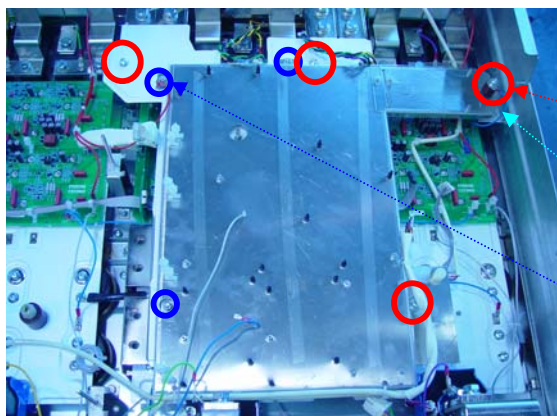


Step 1

Remove 8 screws (J129).

Remove 2 nuts (P1)

Remove 1 screw (P2)



Step 2

Remove 4 screws (P2)

And disconnect one wire:

Blue -> X53 on gate drive board IGBT W

Remove 3 nuts (P1)



Step 3

Remove 6 screws (J129)

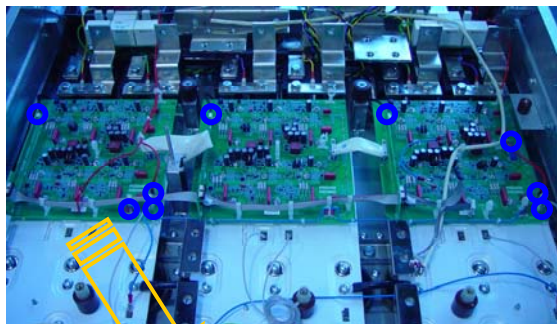
Remove 1 screw (P2)



Mark	Size	Torque
J129	M10x20	27Nm
P2	M6x14	5.5Nm
P1	M6	5.5Nm

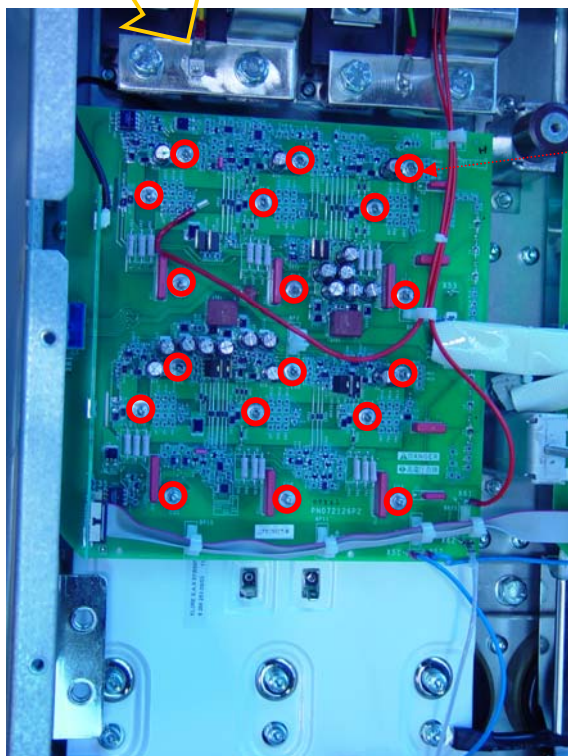
The DC bus bar kit can be removed.

Gate Drive Board IGBT.VX5A1203



Disconnect the 4 wires top to down.

Black X4 on gate drive Board IGBT U/V/W-> thermal sensor
Blue X53 on gate drive board IGBT W-> DC bus
Blue X51 on gate drive Board IGBT U->Power Interconnection bar U
Red X61 on gate drive board U->X10 on snubber rectifier circuit 1
Red X61 on gate drive board W ->X10 on snubber rectifier circuit 2
White X62 on gate drive board U ->discharging resistor
White X62 on gate drive board W ->discharging resistor



For each 3 branch U/V/W, you should remove:

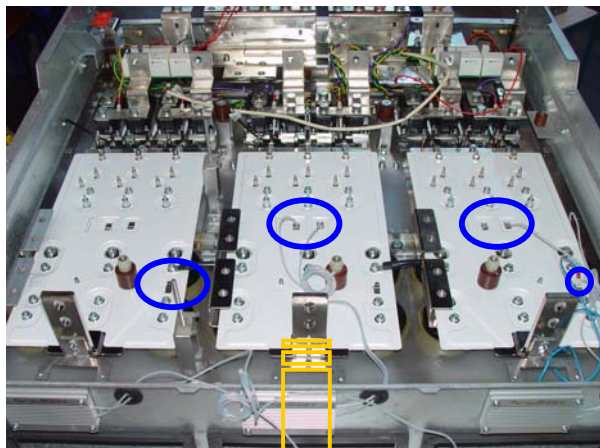
Remove 18 nuts

The Gate Drive Boards IGBT can be removed.



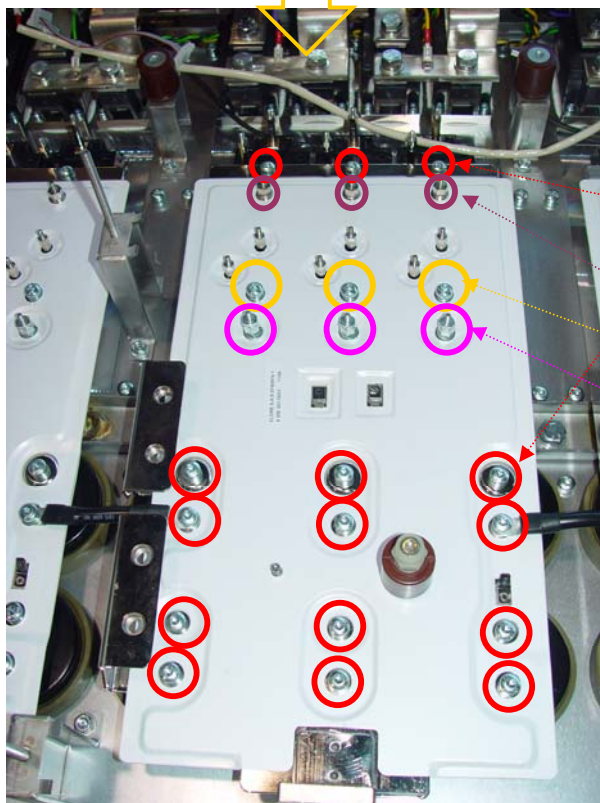
Size	Torque
M4	1.2Nm

DC bus bar kit: VZ3N1339



Disconnect the 4 wires Left to right.

Blue wire on interconnection bar U->X51 on gate drive board IGBT U
White wires on interconnection bar V->discharging resistor
White wires on interconnection bar W->discharging resistor
Blue wire on interconnection bar W ->UD0 on Power Board



For 3 AC bus bar U V, you should remove:

Remove 3 screws (P1).

Remove 12 screws (P2-P3-P4)

Remove 3 screws (P6)

Remove 3 screws (P7)

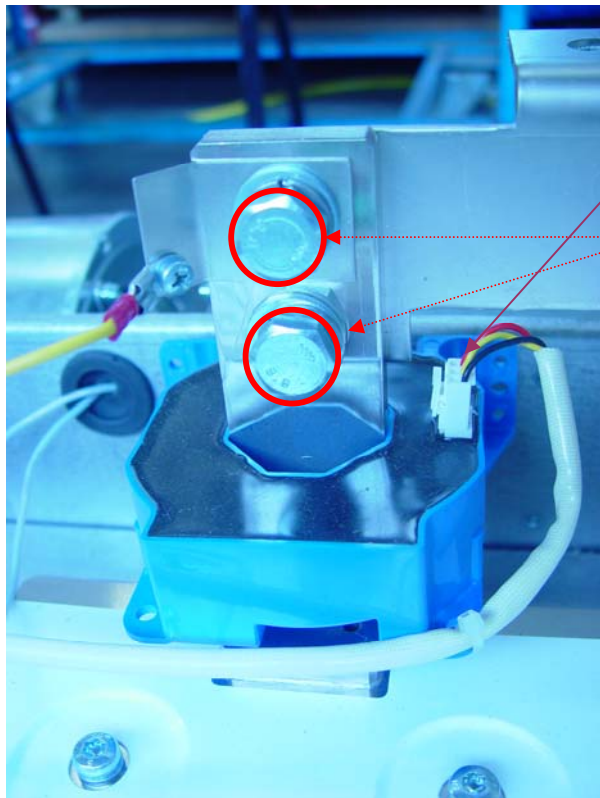
Remove 3 screws (P5)

Be careful, at the time of reassembling;
don't forget to put the 2 Shunt.

The DC Bus Bar KIT can be removed.

Mark	Size	Torque
P1	M6x14	5.5Nm
P2- P3- P4	M6x14	3,3Nm
P6	ST M6/M4	5.5Nm
P7	M6x20	5.5Nm
P5	ST M6/M4	5.5Nm

Motor current sensor: VY1A1109



For 3 current sensor on U/V/W, you should remove:

Disconnect one wire:

Current sensor -> **X11 on measuring board**

Remove 2 screws (P1)



Remove 2 screws (P2)

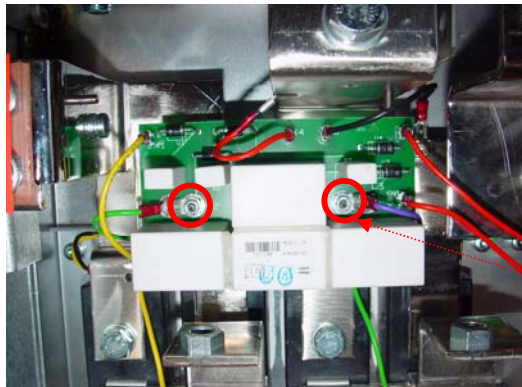
The Motor current sensor can be removed.



Mark	Size	Torque
P1	M12x25	45Nm
P2	M4x12	1.2Nm

Rectifier Snubber Circuit: VX4A1200

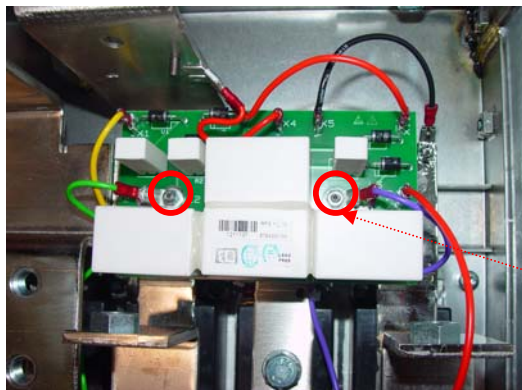
Snubber rectifier circuit 1



Yellow X1 -> bus bar L1.1
 Green X2 -> bus bar L2.1
 Purple X3 -> bus bar L3.1
 Red X10 -> X61 on gate drive board IGBT U
 Red X11 -> RFS31+X11 on power Board
 Black X5 -> bus bar PC
 Red X4 -> bus bar PO.1

Remove 2 nuts (S?)

Snubber rectifier circuit 2



Yellow X1 -> bus bar L1.2
 Green X2 -> bus bar L2.2
 Purple X3 -> bus bar L3.2
 Red X10 -> X61 on gate drive board IGBT W
 Red X11 -> bus bar PO.2
 Black X5 -> bus bar PC
 Red X4 -> bus bar PO.2

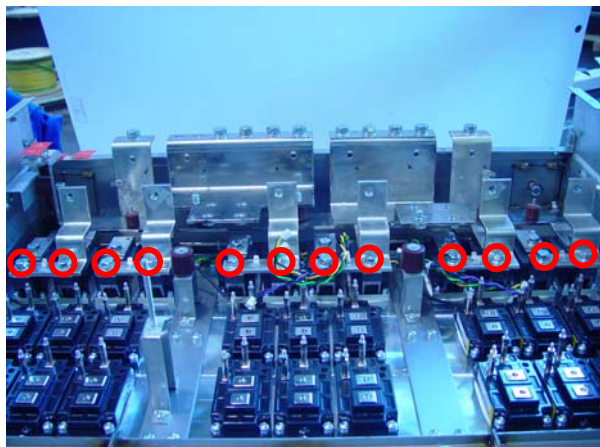
Remove 2 nuts (S?)

The Rectifier Snubber Circuit can be removed.



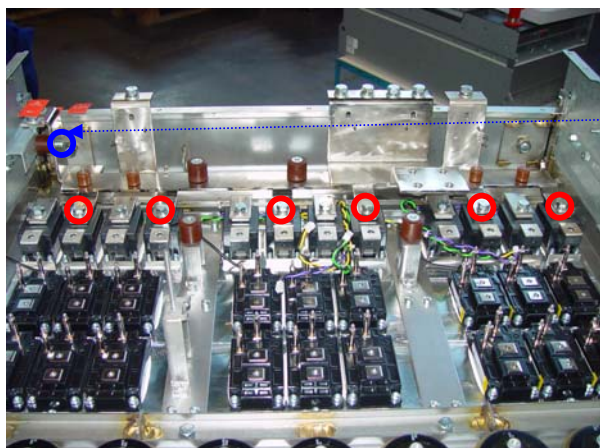
Size	Torque
M4	1.2Nm

Rectifier module thyristor VZ3TM1425M1671
Rectifier module diode VZ3DM1425M1671
AC Bus Bar KIT: VZ3N1340



Part 1

Remove 12 screws (P2)



Part 2

Remove 6 screws (P2)

Remove 1 screw (P1)



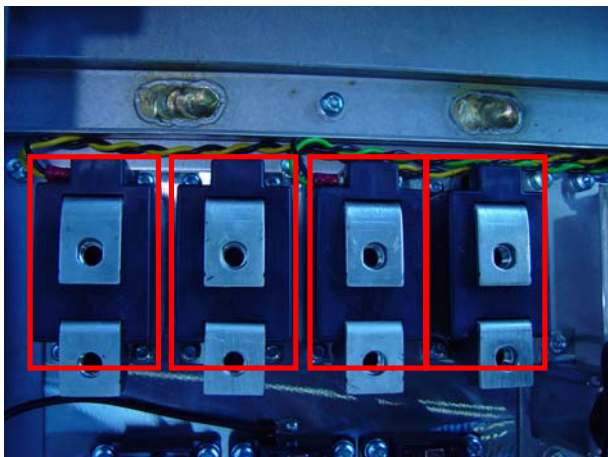
Part 3

Remove 6 screws (P2)



Mark	Size	Torque
P2	M10x25	13.5Nm
P1	M6x14	5.5Nm

Rectifier module thyristor VZ3TM1425M1671 Rectifier module diode VZ3DM1425M1671



For each 3 branch, you should remove:

Remove 16 screws (S?).

It is **not necessary** to change all rectifiers but just this one that is damaged.

After changing, **be careful** to set up direction.
Don't forget applying the grease.



Size	Torque
M5x25	5.0Nm

Rectifier module thyristor VZ3TM1425M1671



Be careful, after changing respect the colour line for the wires.

From left to right

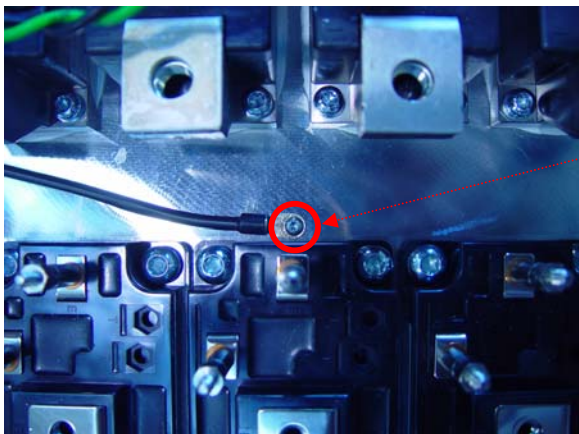
[Yellow-Black]- [Green-Black] - [Purple-Black]

And

[Yellow-Black]- [Green- Black] - [Purple- Black]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Thermal Sensor. VZ3G1104



For each 3 branch U/V/W, you should remove:

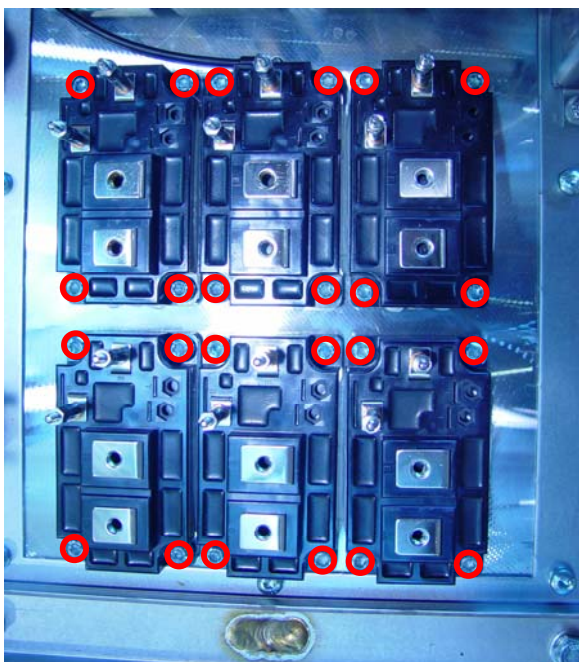
Remove 1 screw

The thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Module IGBT. VZ3IM1603M1271



For each 3 branch U/V/W, you should remove:

Remove 24 screws

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

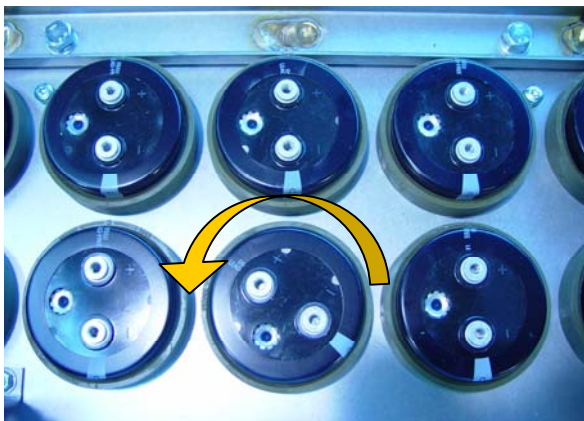
Don't forget applying the grease.

The Module IGBT can be removed.



Size	Torque
M6x20	3.0Nm

Lots of 6 Capacitors. VY1ADC1112



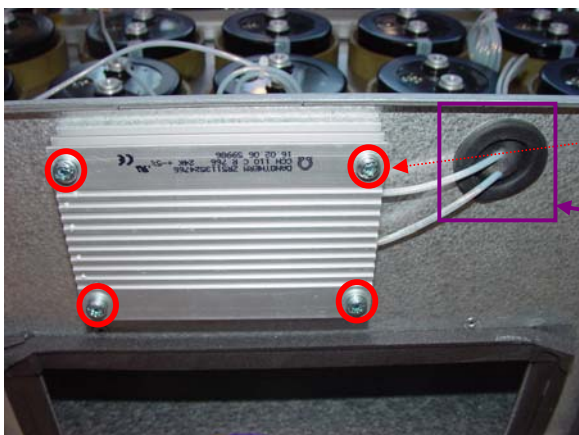
For each 3 branch U/V/W, you should remove 6 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

The Capacitors can be removed.

Discharging Resistor VZ3R24KW125



For each 3 discharging resistor, you should remove:

Remove 4 screws

Be careful, to remove the discharging resistor, remove the gasket and slide the wire in the hole.

The discharging resistor can be removed.



Size	Torque
M4x20	1.2Nm



9.26.2 Product Assembling Drawing

No information

9.26.3 Product Cabling Drawing

Refer to following file: [circuit diagram 14A.pdf](#)

Refer to following file: [circuit diagram 14B.pdf](#)



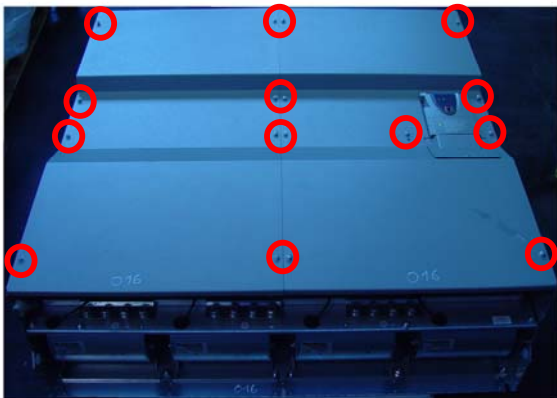
9.27 ATV61/71 Size 15 (size, refer to 1.2)

9.27.1 Dismantling and reassembling

Size 15: ATV71HC50N4, ATV61HC63N4

ATV71HC50N4		ATV61HC63N4	
Reference	Designation	Reference	Designation
VZ3V1213	Internal Fan	VZ3V1213	Internal Fan
VZ3V1212	Fan Power Electronic	VZ3V1212	Fan Power Electronic
VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)	VZ3TM1600M1671	Rectifier Module Thyristor (600A / 1600V)
VZ3R24KW125	Discharging Resistor (24k / 125W)	VZ3R24KW125	Discharging Resistor (24k / 125W)
VZ3N1334	Wires KIT	VZ3N1334	Wires KIT
VZ3N1332	AC Bus Bar KIT	VZ3N1332	AC Bus Bar KIT
VZ3N1330	DC Bus Bar KIT	VZ3N1330	DC Bus Bar KIT
VZ3IM1604M1271	Lot of 4 Modules Paired IGBT (600A / 1200V)	VZ3IM1604M1271	Lot of 4 Modules Paired IGBT (600A / 1200V)
VZ3G1104	Thermal Sensor	VZ3G1104	Thermal Sensor
VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)	VZ3DM1600M1671	Rectifier Module Diode (600A / 1600V)
VY1ADV1111	Screws KIT	VY1ADV1111	Screws KIT
VY1ADC1112	Lots of 6 capacitors (5200µF / 400V)	VY1ADC1112	Lots of 6 capacitors (5200µF / 400V)
VY1A1407	Plastic Parts KIT	VY1A1407	Plastic Parts KIT
VY1A1306	Assembling KIT	VY1A1306	Assembling KIT
VY1A1217	Front Cover with I/O Terminal Cover	VY1A1217	Front Cover with I/O Terminal Cover
VY1A1109	Motor Current Sensor	VY1A1109	Motor Current Sensor
VX5A1HC5063	Power Board	VX5A1HC5063	Power Board
VZ3F1113	Braking Unit KIT	VZ3F1113	Braking Unit KIT
VX5A1400	Fan Control Board	VX5A1400	Fan Control Board
VX5A1300	Soft Charge Board	VX5A1300	Soft Charge Board
VX5A1204	Gate Drive Board	VX5A1204	Gate Drive Board
VX4A71101Y	Control bloc P >=90kW	VX4A61101Y	Control bloc P >=90kW
VX4A1200	Rectifier snubber circuit	VX4A1200	Rectifier snubber circuit
VX4A1117	RFI Filter Board	VX4A1117	RFI Filter Board
VW3A4526	DC Choke	VW3A4526	DC Choke
VX4A1104	Terminal Board	VX4A1104	Terminal Board
VX4A1103	Front cover 4x7 Digits	VX4A1103	Front cover 4x7 Digits

Front Cover with I/O Terminal Cover: VY1A1217

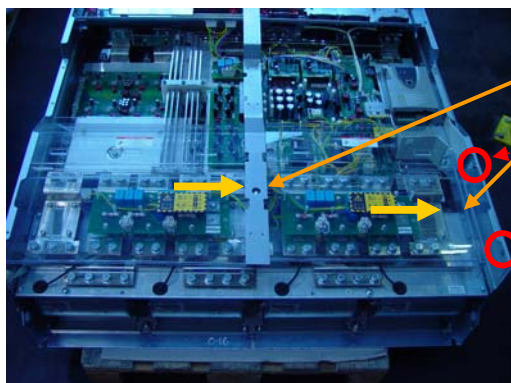


Remove 17 screws



Size	Torque
M6x12	5.5Nm

Plastic Parts KIT: VY1A1407



Push here and disengage plastic part

Remove 2 nuts



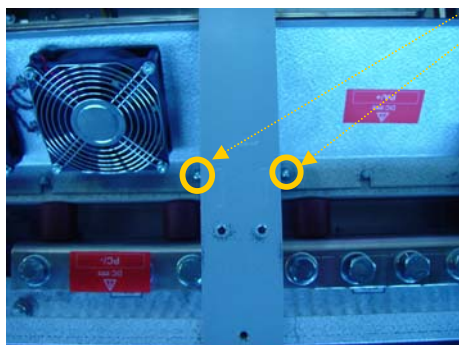
Size	Torque
M6	5.5Nm

The Plastic part can be removed.

Metal kit.



Remove 3 screws (P1)



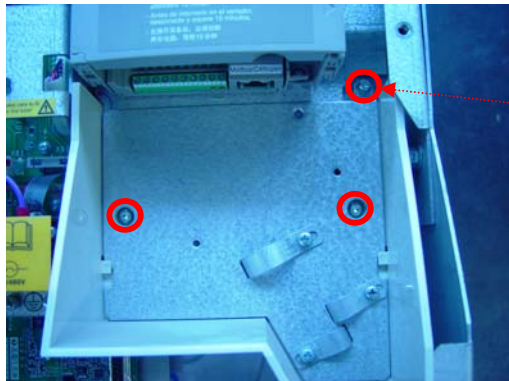
Remove 2 nuts (P2)



The metal part can be removed.

Mark	Size	Torque
P1	M6x14	5.5Nm
P2	M6	5.5Nm

Plastic Parts KIT: VY1A1407



Remove 3 screws

The Plastic Part kit can be removed.



Size	Torque
M4x12	1.2Nm

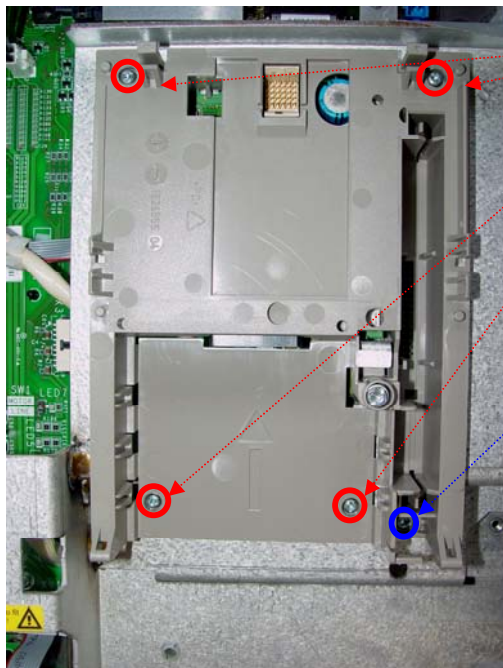
Control bloc: VX4A71101Y



For remove the terminal board, turn the screw in the unbolt position and pulls forward to take out terminal board.



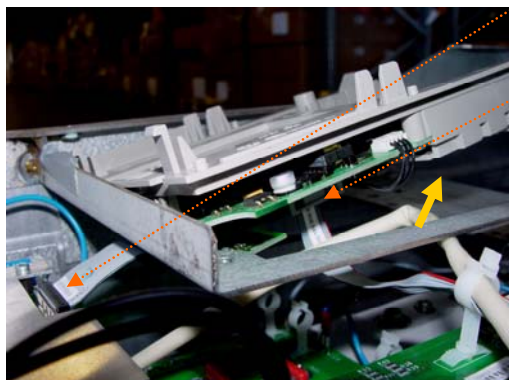
Press the two clips and pulls forward to take out the display board.



Remove 4 screws (S5 to S8).

Remove 1 screw (S38).

Disconnect the ribbon cable.
X3 from Control bloc "interface Board"->X3 on Motor Control Board



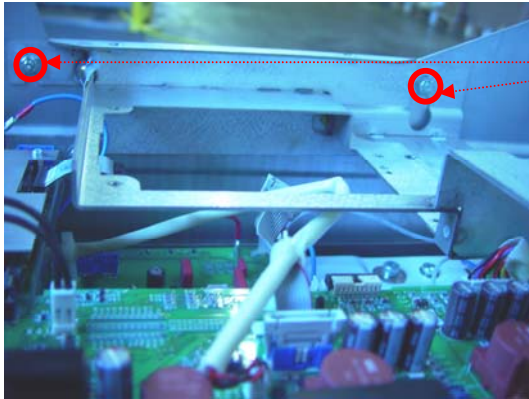
Disconnect the ribbon cable.
X4 from Control bloc" interface Board"->X4 on Power Board

The Control bloc can be removed.

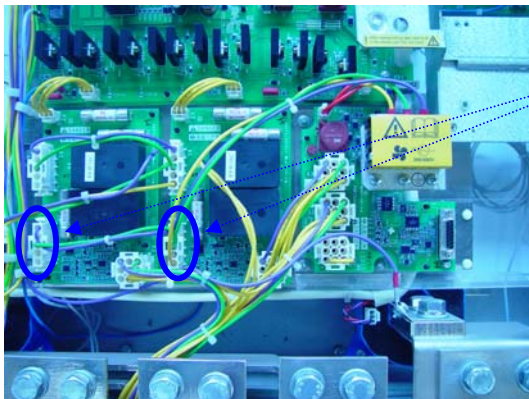


Mark	Size	Torque
S5-S8	M3x12	0.78Nm
S38	M4x6	1.5NM

Metal Support for Control Bloc



Remove 2 nuts



Disconnect wire:

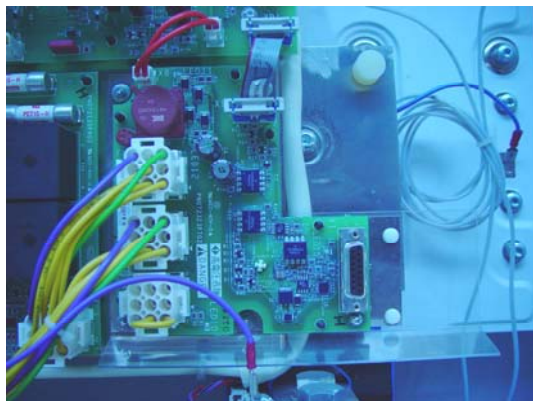
Yellow/Green/Purple TB1 -> X4 on both fan control board

The metal kit can be removed.



Size	Torque
M6	5.5Nm

Braking Unit KIT: VZ3F1113



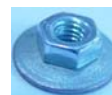
Disconnect 5 wires top to down:

- Red X8 -> X81 on Power board
- Ribbon X91 -> X91 on Power board
- Yellow/Green/Purple X3 -> X3 on fan control board 2
- Yellow/Green/Purple X33 -> X13 & X14 on RFI filter board 2
- Yellow X2 -> Shunt X2 on braking unit device



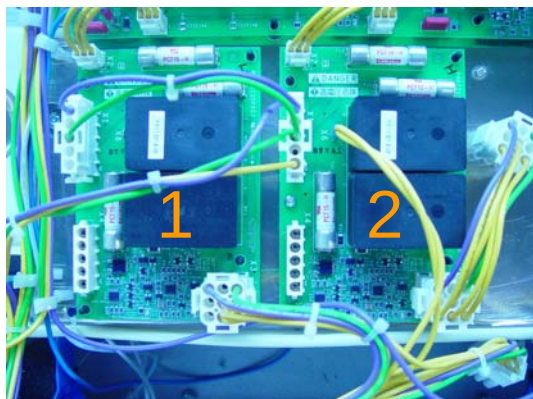
Remove 4 nuts

The Braking unit device can be removed



Size	Torque
M3	0.8Nm

Fan Control Board: VX5A1400



Disconnect wires. From top to down:

Fan control Board 1

Yellow X2->X14 on Power board

Yellow/Green/Purple X1->X1 on fan control Board 2

Yellow/Green/Purple X3-> X11 & X12 on RFI filter board 1

Fan control Board 2

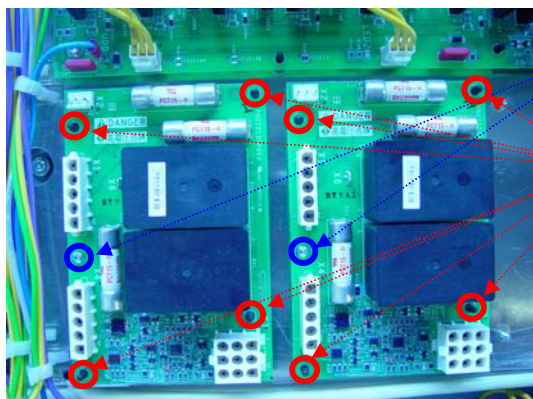
Yellow X2->X15 on Power board

Yellow/Green/Purple X1->X1 on Fan control board 1

Yellow X1/Green X2/ Purple X3 on RFI filter board 1

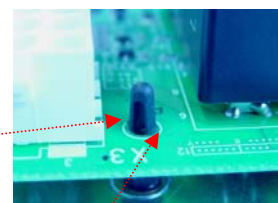
Yellow X6-> X13 & X14 on RFI filter board 2

Yellow/Green/Purple X3-> X3 on braking unit device



Remove 2 screws

Push on the 8 plastic supports to remove the board.



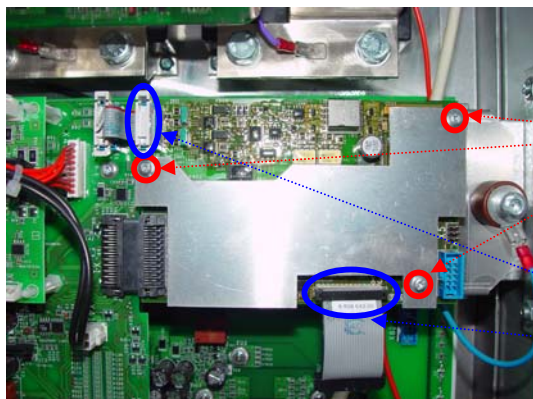
Push on the side.



Size	Torque
M3x6	0.8Nm
M3x8	clips

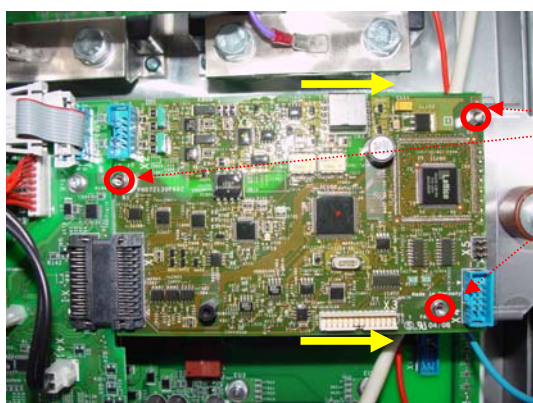
The Fan control board can be removed.

Motor control Board: VX4A71101Y



Remove the 3 screws. (P1)

Disconnect the 2 ribbon cable, *left to the right*:
X2 -> X2 on Power board
X3 -> X3 on Interface Board in Control Block



Remove the 3 hexagonal studs (P2).

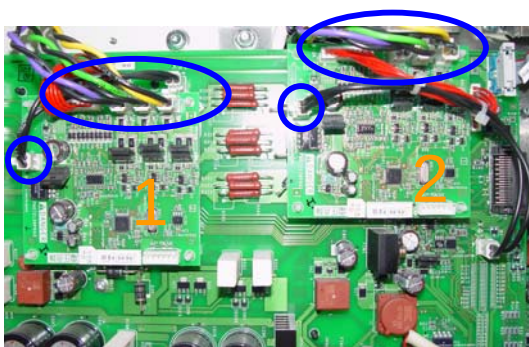
Be careful, at the time of reassembling;
don't forget to put the steel.



Mark	Size	Torque
P1	M3x6	0.8Nm
P2	M3x10	0.8Nm

The motor control board can be removed.

Soft Charge board: VX5A1300



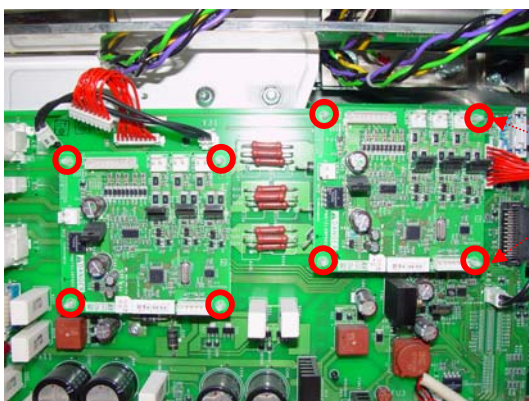
Disconnect wires, *left to the right*:

Soft charge Board 1

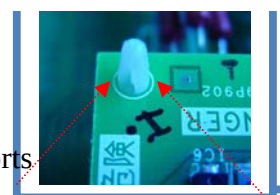
Blue CN7A-> X31 on Power board
Red CN2A-> X30 on power board
Purple/Black CNL3G ->Gate rectifier 3.1 on L3.1
Green/Black CNL2G ->Gate rectifier 2.1 on L2.1
Yellow/Black CNL1G -> Gate rectifier 1.1 on L1.1

Soft charge Board 2

Blue CN7A-> X41 on Power board
Red CN2A-> X40 on power board
Purple/White CNL3G ->Gate rectifier 3.2 on L3.2
Green/ White CNL2G ->Gate rectifier 2.2 on L2.2
Yellow/ White CNL1G -> Gate rectifier 1.2 on L1.2



Push on the 8 plastic supports
to remove the board.



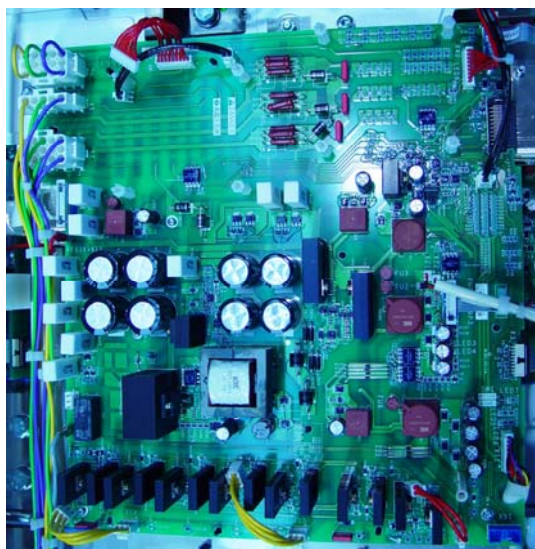
Push on the side.



Size	Torque
M3x8	clips

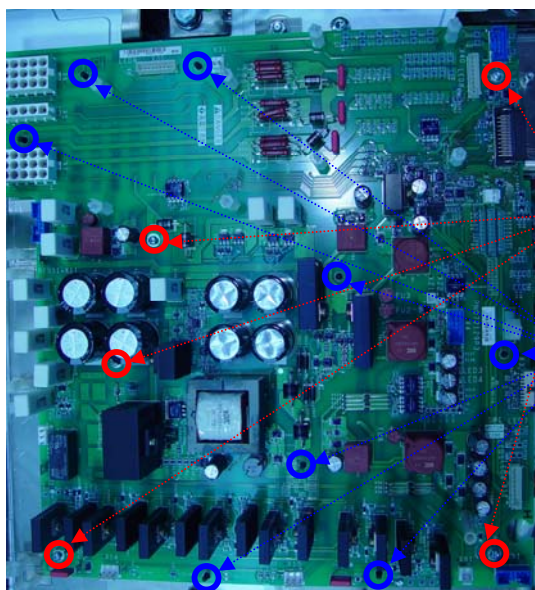
The Soft charge Board can be removed.

Power Board: VX5A1HC5063



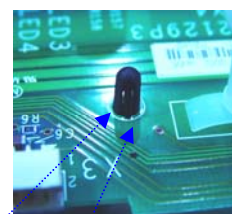
Disconnect the 21 wires, *unclockwise*:

Black X41-> CN7A on soft charge board 2
 Red X40-> CN2A on soft charge board 2
 Ribbon X2-> X2 on motor control board
 Black X31-> CN7A on soft charge board 1
 Red X30-> CN2A on Soft charge board 1
 Yellow/Green/Purple X6-> X6 shunt on Power Board
 Yellow/Green/Purple X5-> bus bar U Yellow, bus bar V Green, bus bar W Purple
 Yellow/Green/Purple X7->Yellow X11/Green X12/Purple X13 on both RFI filter Board
 Ribbon X8 -> X82 on Gate drive Board IGBT U
 Red RFS31+X11 -> X11 on Snubber circuit 1
 White UD-R1 -> discharging resistor
 Blue UD0 -> interconnection bus bar W
 Yellow X14-> X2 on Fan control Board 1
 Yellow X15 -> X2 on Fan control board 2
 Red X81-> X8 on Braking unit device
 Ribbon X91-> X91 on Braking unit device
 Multicolour X11-> currents sensors U V W
 Ribbon X3-> X32 on Gate drive board W
 Ribbon X4-> X4 on control block
 White X22-> inside Fan alimentation
 White X21-> inside Fan alimentation



Remove the 5 screws. (S?)

Push on the 8 plastic supports to remove the board



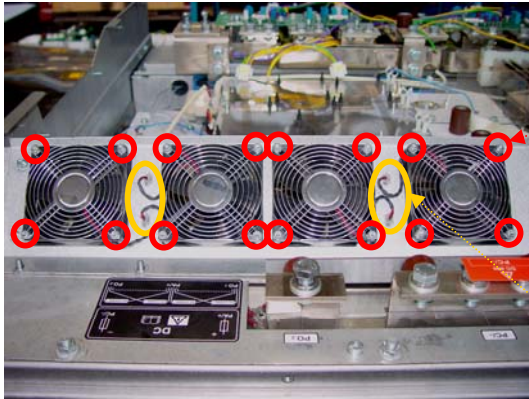
Push on the side.



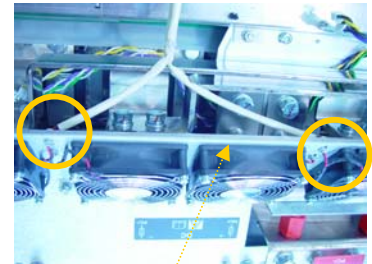
Size	Torque
M3x6	0.8Nm
M3x8	clips

The Power board can be removed.

Internal Fan: VZ3V1213



Remove 16 screws.



Disconnect 4 Wires

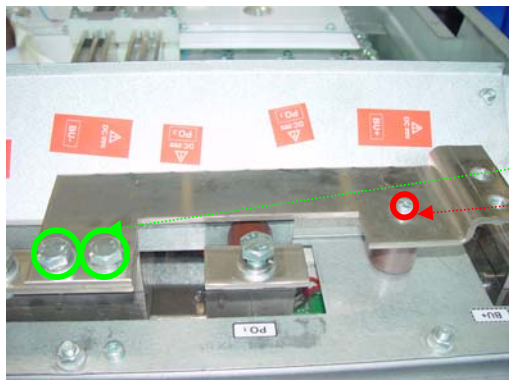
Black Fan-> X21 X22 on Power board

The Fan can be removed.



Size	Torque
M4x35	1.2Nm

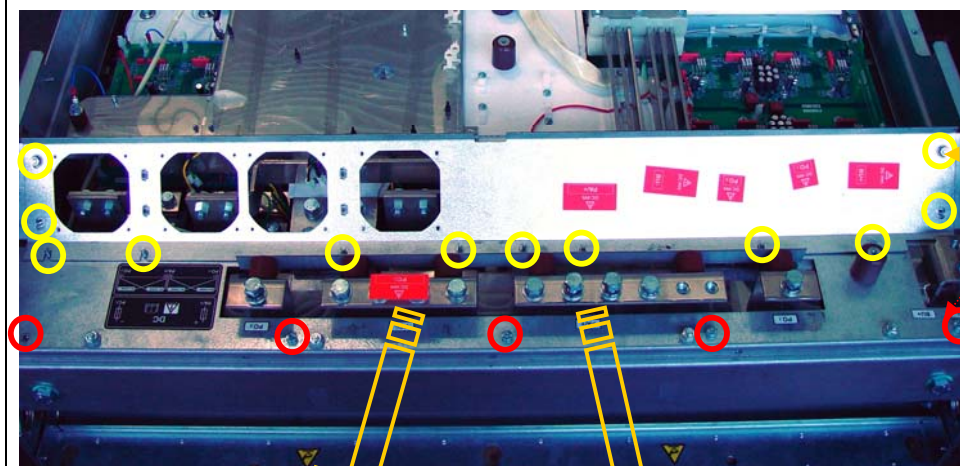
AC Bus Bar KIT: VZ3N1332



Step 1

Remove 2 screws (P3).

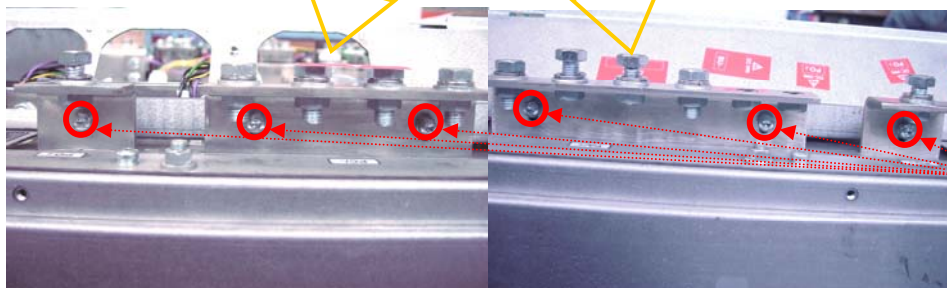
Remove 1 screw (P1)



Step 2

Remove 12 nuts (P2).

Remove 5 screws (P1)



Step 3

Remove 6 screws (P1).

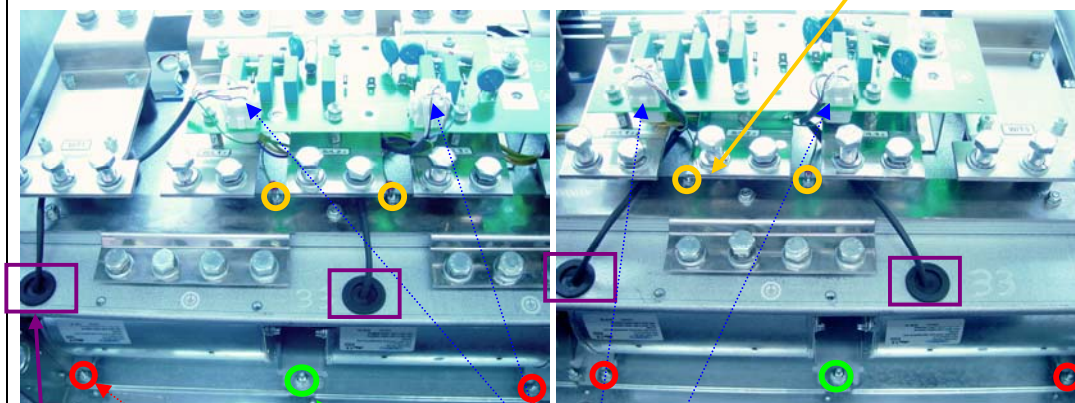


Mark	Size	Torque
P3	M12x25	45Nm
P1	M6x14	5.5Nm
P2	M6	5.5Nm

The AC Bus Bar Kit can be removed.

Fan Power Electronic: VZ3V1212

Be careful; don't forget to disconnect the ground wire.



Disconnect 4 connectors

Fan turbine 1->X11 on RFI filter board 1->X3 on fan control board 1

Fan turbine 2->X12 on RFI filter board 1->X3 on fan control board 1

Fan turbine 3->X13 on RFI filter board 2->X33 on Braking unit device

Fan turbine 4->X14 on RFI filter board 2->X33 on Braking unit device

Remove 4 screws (J130)

Remove 2 screws (J139)

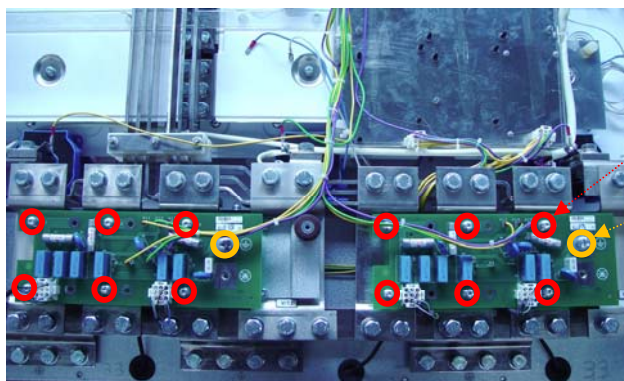
Be careful, to remove the fan, remove the gasket and slide the wire in the hole.

The Turbine can be removed.



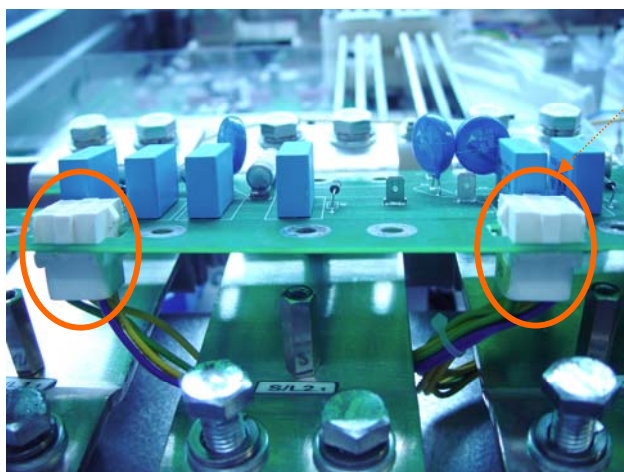
Mark	Size	Torque
J130	M6	5.5Nm
J139	M8	13.5Nm

RFI Filter Board: VX4A1117



Remove 12 screws (P1)

Remove 2 screws (P2)



To change the board, don't forget to disconnect the 4 connectors

X11 X12 on RFI Filter Board 1

And

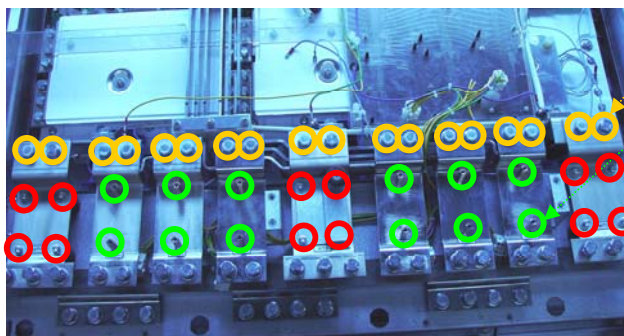
X13 X14 on RFI Filter Board 2

The Both RFI filter board can be removed.



Mark	Size	Torque
P2	M6x12	5.5Nm
P1	M6x14	5.5Nm

AC Bus Bar KIT: VZ3N1332



Remove 18 screws (P1)

Remove 12 screws (P2)

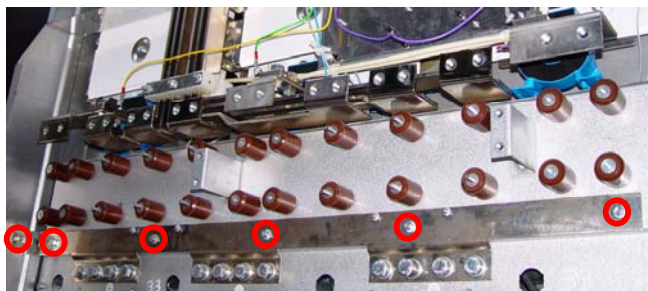
Remove 12 screws (P3)



Mark	Size	Torque
P1	M12x25	45Nm
P2	M6x22	5.5Nm
P3	M6x14	5.5Nm

The AC Bus Bar kit can be removed.

Earth Terminal Bar: VZ3N1332



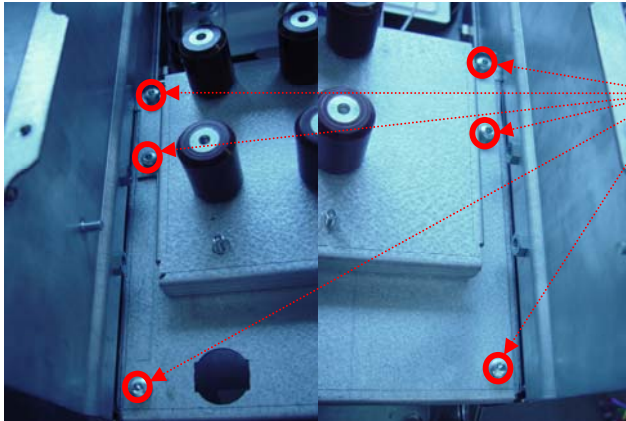
Remove 6 nuts



Size	Torque
M6	5.5Nm

The Earth Terminal Bar can be removed.

Metal Body Part KIT



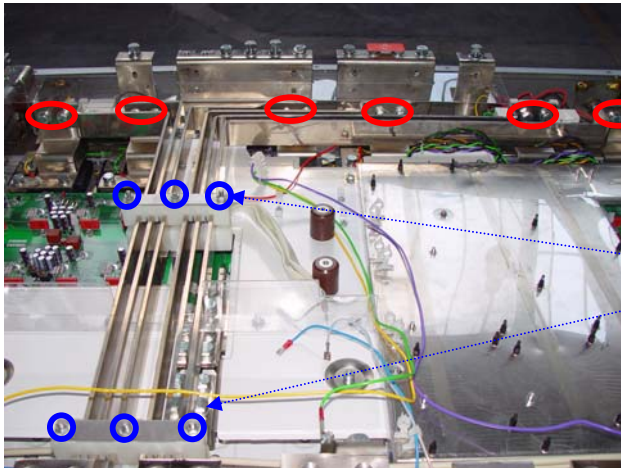
Remove 6 screws



Size	Torque
M6x14	5.5Nm

The metal kit can be removed.

AC Bus Bar KIT: VZ3N1332



Remove 12 screws

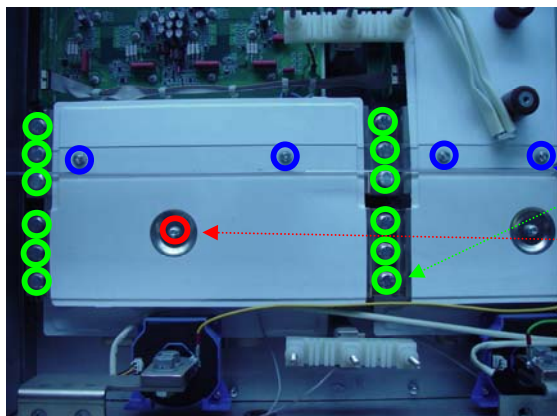
Remove 6 nuts



Size	Torque
M12x25	45Nm
M6	5.5Nm

The AC Bus bar kit can be removed.

DC Bus Bar KIT: VZ3N1330

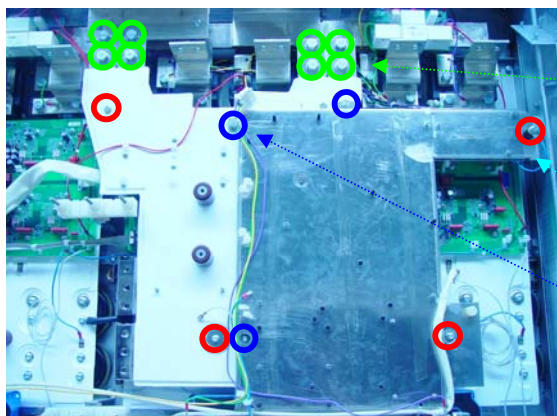


Step 1

Remove 12 screws (J129).

Remove 4 nuts (P1)

Remove 1 screw (P2)



Step 2

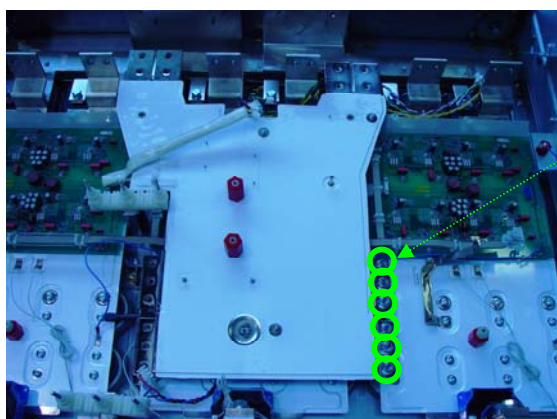
Remove 8 screws (J129).

Remove 5 screws (P2)

And disconnect one wire:

Blue -> X53 on gate drive board IGBT W

Remove 3 nuts (P1)



Step 3

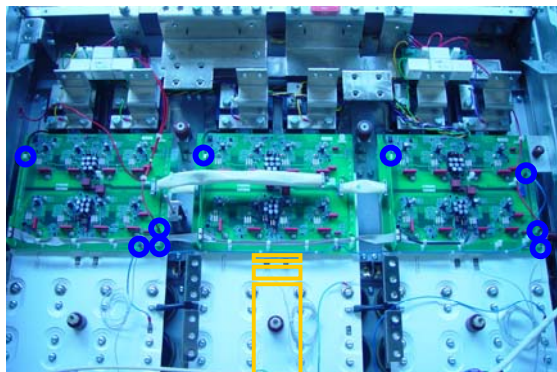
Remove 6 screws (J129)

The DC bus bar kit can be removed.



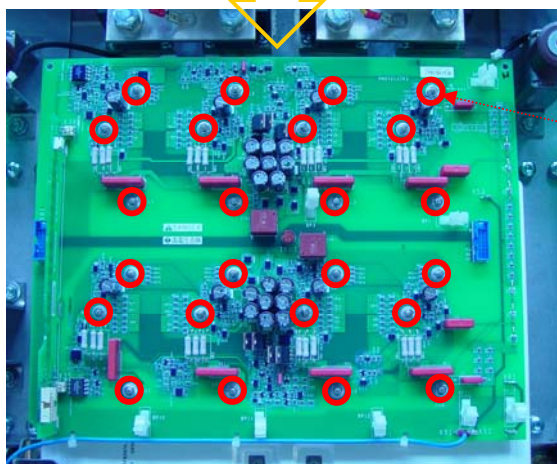
Mark	Size	Torque
J129	M10x20	27Nm
P2	M6x14	5.5Nm
P1	M6	5.5Nm

Gate Drive Board IGBT.VX5A1204



Disconnect the 4 wires top to down.

Black X4 on gate drive Board IGBT U/V/W-> thermal sensor
Blue X53 on gate drive board IGBT W-> DC bus
Blue X51 on gate drive Board IGBT U->Power Interconnection bar U
Red X61 on gate drive board U->X10 on snubber rectifier circuit 1
Red X61 on gate drive board W ->X10 on snubber rectifier circuit 2
White X62 on gate drive board U ->discharging resistor
White X62 on gate drive board W ->discharging resistor



For each 3 branch U/V/W, you should remove:

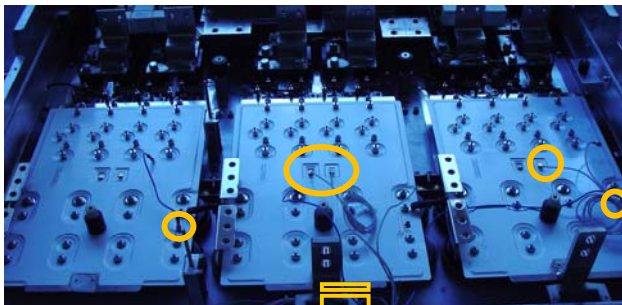
Remove 24 nuts

The Gate Drive Boards IGBT can be removed.



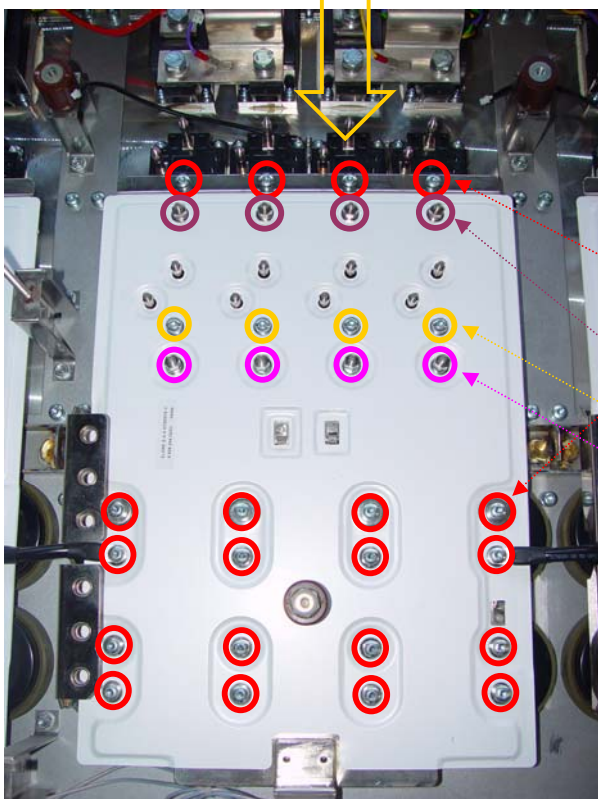
Size	Torque
M4	1.2Nm

DC Bus Bar KIT: VZ3N1330



Disconnect the 4 wires Left to right.

Blue wire on interconnection bar U->X51 on gate drive board IGBT U
White wires on interconnection bar V->discharging resistor
White wires on interconnection bar W->discharging resistor
Blue wire on interconnection bar W ->UD0 on Power Board



For 3 AC bus bar U V, you should remove:

Remove 4 screws (P1)

Remove 16 screws (P2-P3-P4).

Remove 4 screws (P6)

Remove 4 screws (P7)

Remove 4 screws (P5)

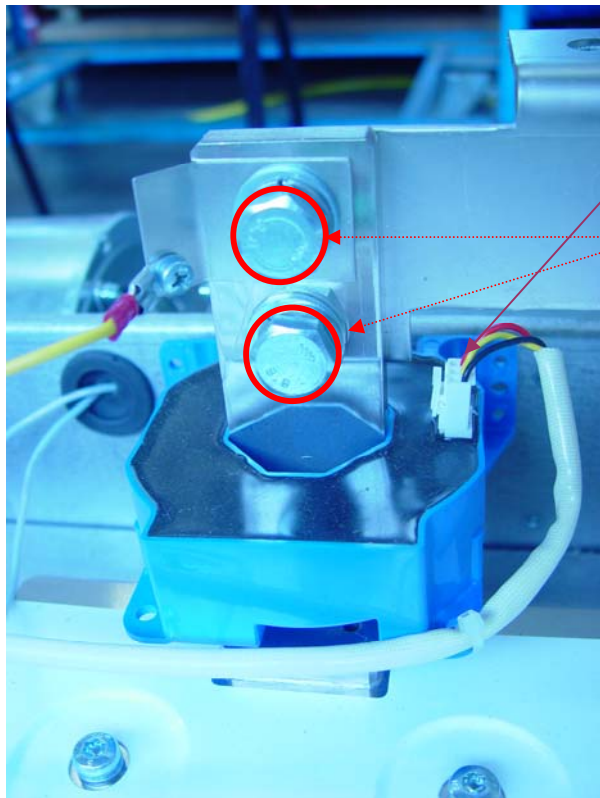
Be careful, at the time of reassembling, don't forget to put the 2 Shunt.

The DC Bus Bar-Phase can be removed.



Mark	Size	Torque
P1	M6x14	5.5Nm
P2-P3-P4	M6x14	3,3Nm
P6	ST M6/M4	5.5Nm
P7	M6x20	5.5Nm
P5	ST M6/M4	5.5Nm

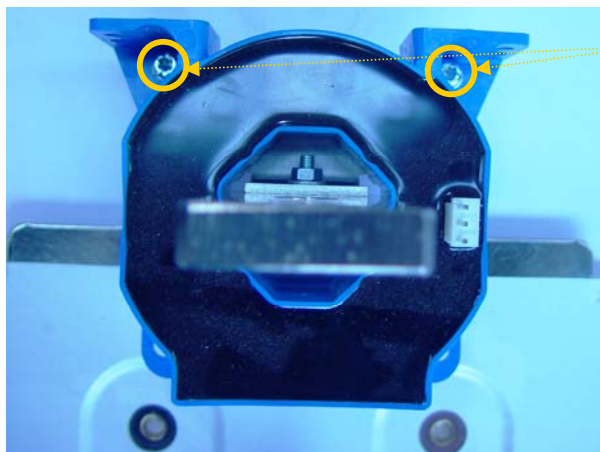
Motor Current Sensor: VY1A1109



For 3 current sensor on U/V/W, you should remove:

Disconnect one wire:
Current sensor -> X11 on measuring board

Remove 2 screws (P1)



Remove 2 screws (P2)

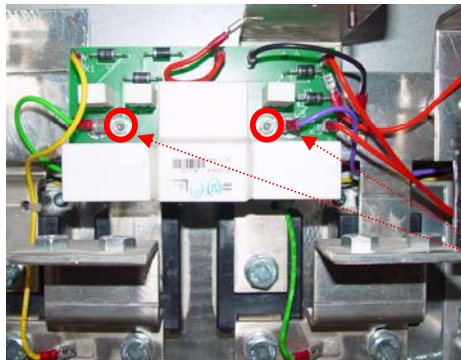
The current sensor can be removed.



Mark	Size	Torque
P1	M12x25	45Nm
P2	M4x12	1.2Nm

Rectifier Snubber Circuit: VX4A1200

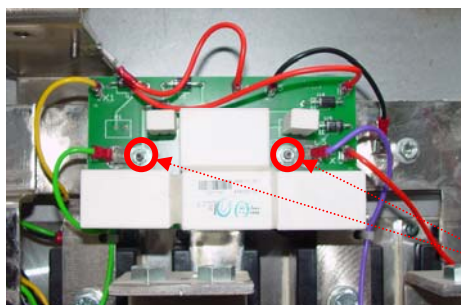
Snubber rectifier circuit 1



Yellow X1 -> bus bar L1.1
Green X2 -> bus bar L2.1
Purple X3 -> bus bar L3.1
Red X10 -> X61 on gate drive board IGBT U
Red X11 -> RFS31+X11 on power Board
Black X5 -> bus bar PC
Red X4 -> bus bar PO.1

Remove 2 nuts

Snubber rectifier circuit 2



Yellow X1 -> bus bar L1.2
Green X2 -> bus bar L2.2
Purple X3 -> bus bar L3.2
Red X10 -> X61 on gate drive board IGBT W
Red X11 -> bus bar PO.2
Black X5 -> bus bar PC
Red X4 -> bus bar PO.2

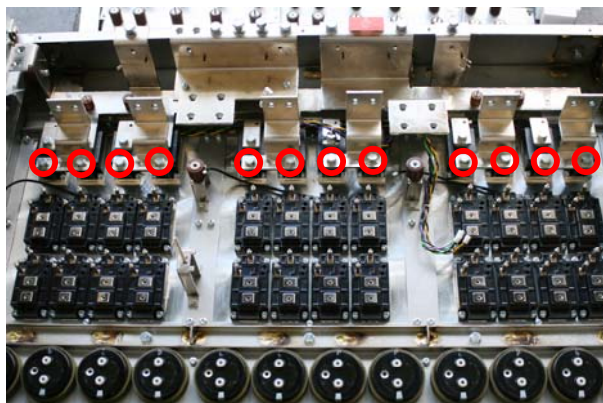
Remove 2 nuts

The Rectifier Snubber Circuit can be removed.



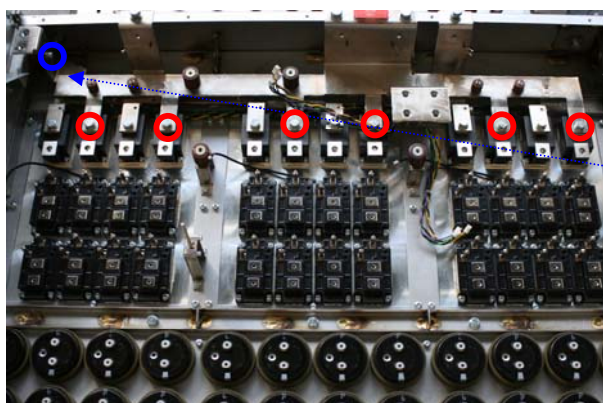
Size	Torque
M4	1.2Nm

Rectifier Module Thyristor: VZ3TM1600M1671
Rectifier Module Diode: VZ3DM1600M1671
AC Bus Bar KIT: VZ3N1332



Part 1

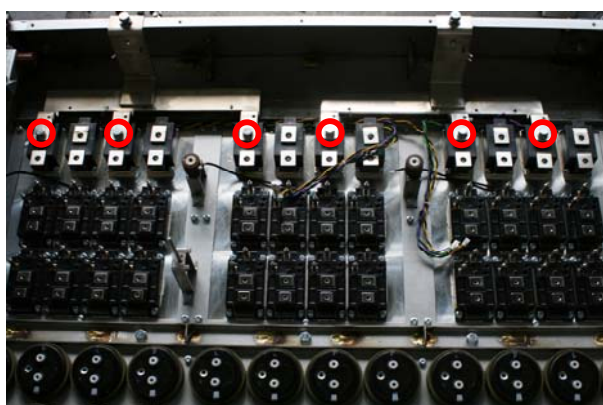
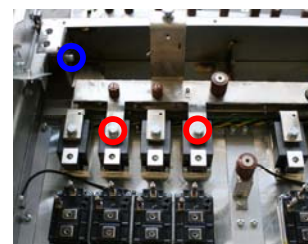
Remove 12 screws (P2)



Part 2

Remove 6 screws (P2)

Remove 1 screw (P1)



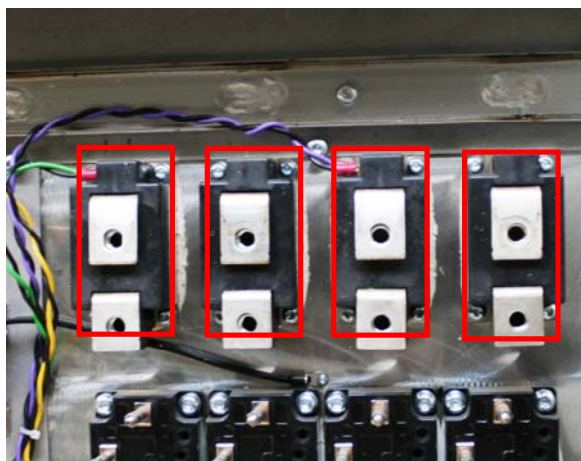
Part 3

Remove 6 screws (P2)



Mark	Size	Torque
P2	M10x25	13.5Nm
P1	M6x14	5.5Nm

Rectifier Module Thyristor: VZ3TM1600M1671 Rectifier Module Diode: VZ3DM1600M1671



For each 3 branch, you should remove:

Remove 16 screws

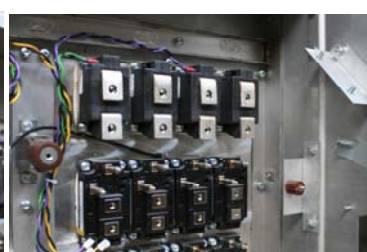
It is **not necessary** to change all rectifiers but just this one that is damaged.

After changing, **be careful** to set up direction.
Don't forget applying the grease.



Size	Torque
M5x25	5.0Nm

Rectifier Module Thyristor: VZ3TM1600M1671



Be careful, after changing respect the colour line for the wires.

From left to right

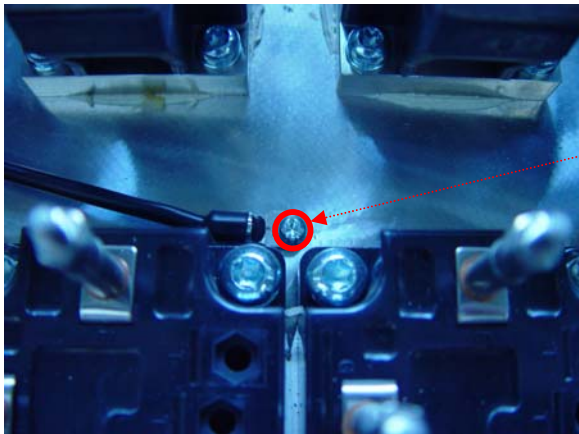
[Yellow-Black]- [Green-Black] - [Purple-Black]

And

[Yellow-White]- [Green- White] - [Purple- White]

It is **not necessary** to change all rectifiers but just this one that is damaged.

Thermal Sensor: VZ3G1104



For each 3 branch U/V/W, you should remove:

Remove 1 screw

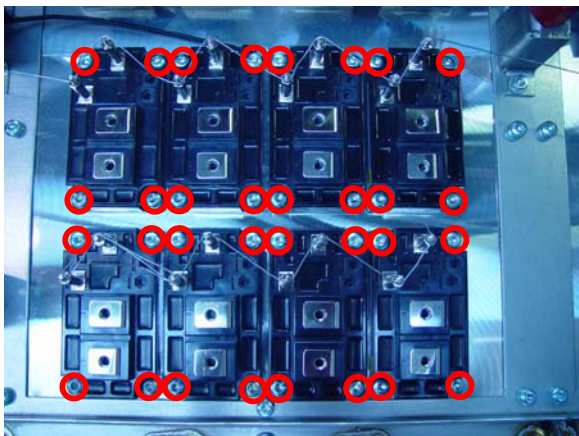
Be careful, don't forget to connect the wire on the connector "X4" of the Gate drive board IGBT

The thermal sensor can be removed.



Size	Torque
M3x6	0.8Nm

Lot of 4 Modules Paired IGBT: VZ3IM1604M1271



For each 3 branch U/V/W, you should remove:

Remove 32 screws

It is **not necessary** to change all IGBT but just this one that is damaged.

After changing, **be careful** to set up direction.

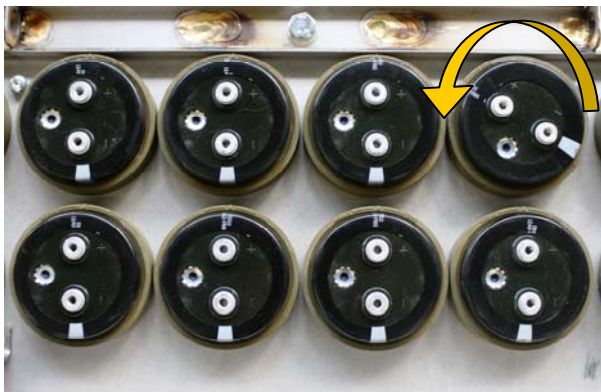
Don't forget applying the grease.

The IGBT can be removed.



Size	Torque
M6x20	3.0Nm

Lots of 6 capacitors: VY1ADC1112



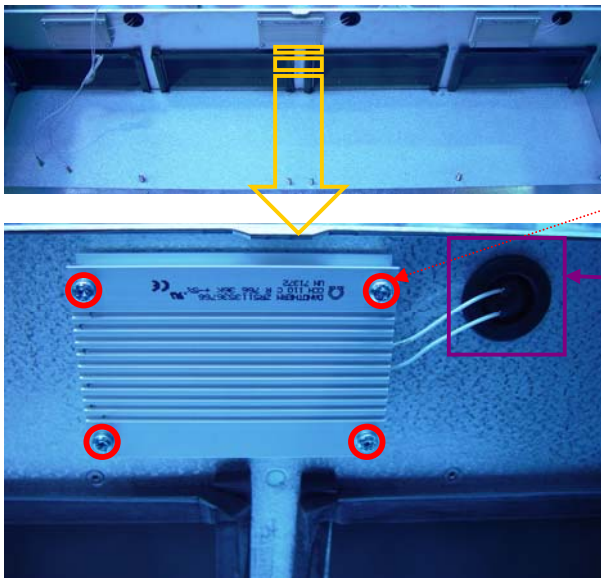
For each 3 branch U/V/W, you should remove 6 capacitors:

To remove the capacitor, rotate left it, and pull out it

After changing, **be careful** to set up direction

The Capacitors can be removed.

Discharging Resistor: VZ3R24KW125



For each 3 discharging resistor, you should remove:

Remove 4 screws

Be careful, to remove the discharging resistor, remove the gasket and slide the wire in the hole.

The discharging resistor can be removed.



Size	Torque
M4x20	1.2Nm



9.27.2 Product Assembling Drawing

No information

9.27.3 Product Cabling Drawing

Refer to following file: [circuit diagram 15.pdf](#)

Soon available

9.29 ATV61/71 Size 13V (size, refer to 1.2)

9.29.1 Dismantling and reassembling

Size 13V: ATV71HC20Y, ATV71HC25Y, ATV71HC31Y, ATV61HC25Y, ATV61HC31Y, ATV61HC40Y

ATV71HC20Y	
Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1348	Wires KIT
VZ3N1345	AC Bus Bar KIT
VZ3N1342	DC Bus Bar KIT
VZ3IM1502M1771	Lot of 2 IGBT Modules Related (500A / 1700V)
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100μF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC200250	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1120	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX5A1400	Fan Control Board
VZ3V1213	Internal Fan
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VZ3F1112	Braking Unit KIT
VX4A1103	Front cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HC25Y	
Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1348	Wires KIT
VZ3N1345	AC Bus Bar KIT
VZ3N1342	DC Bus Bar KIT
VZ3IM1502M1771	Lot of 2 IGBT Modules Related (500A / 1700V)
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100μF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC250315	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1120	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX5A1400	Fan Control Board
VZ3V1213	Internal Fan
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VZ3F1112	Braking Unit KIT
VX4A1103	Front cover 4x7 Digits
VX4A1104	Terminal Board

ATV71HC31Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1348	Wires KIT
VZ3N1345	AC Bus Bar KIT
VZ3N1342	DC Bus Bar KIT
VZ3IM1502M1771	Lot of 2 IGBT Modules Related (500A / 1700V)
VZ3G1105	Thermal Sensor
VZ3F1112	Braking Unit KIT
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC315400	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A71101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1120	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VZ3V1213	Internal Fan
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VX4A1103	Front cover 4x7 Digits

ATV61HC25Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1348	Wires KIT
VZ3N1345	AC Bus Bar KIT
VZ3N1342	DC Bus Bar KIT
VZ3IM1502M1771	Lot of 2 IGBT Modules Related (500A / 1700V)
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC200250	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1120	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VZ3F1112	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

ATV61HC31Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
VZ3N1348	Wires KIT
VZ3N1345	AC Bus Bar KIT
VZ3N1342	DC Bus Bar KIT
VZ3IM1502M1771	Lot of 2 IGBT Modules Related (500A / 1700V)
VZ3G1105	Thermal Sensor
VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC250315	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
VX4A1206	Rectifier Snubber Circuit
VX4A1120	RFI Filter Board
VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VZ3F1112	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

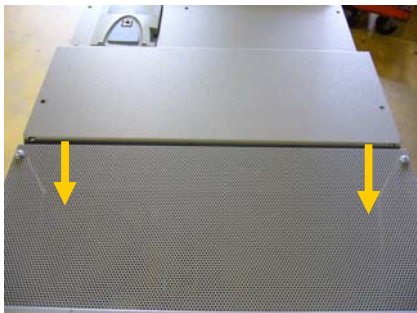
ATV61HC31Y

Reference	Designation
VZ3TM2200M4371	Rectifier Module Thyristor (430A / 2200V)
VZ3R36KW125	Discharge Resistor (36k / 125W)
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VZ3DM2200M5471	Rectifier Module Diode (540A / 2200V)
VY1ADV1113	Screws KIT
VY1ADC1117	Lot of 6 Capacitors (550V / 2100µF)
VY1A1410	Plastic Parts KIT
VY1A1308	Assembling KIT
VX5A71HC250315	Measuring Board
VX5A1HC2540	Power Board
VX5A1301	Soft Charge Board
VX5A1206	Gate Drive Board
VX5A1105	Active Balancing Board
VX4A61101Y	Control bloc P >=90kW
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VZ3V1212	Fan Power Electronic
VX4A1104	Terminal Board
VX5A1400	Fan Control Board
VY1A1108	Motor Current Sensor
VY1A1214	Front Cover with I/O Terminal Cover
VZ3F1112	Braking Unit KIT
VZ3V1213	Internal Fan
VX4A1103	Front cover 4x7 Digits

Front Cover with I/O terminal cover: VY1A1214



Remove 9 screws.



The front cover can be removed.



Size	Torque
M6x12	5.5Nm